

ORIGINAL ARTICLE

Global Production Automation and China's Export Adjustment—Theoretical and Empirical Evidence

Huan Ma¹ | Guang Yang²  | Yanyan Zheng³¹School of International Trade and Economics, Shanghai Cooperation Organization Research Center, Dongbei University of Finance and Economics, Dalian, China | ²School of Economics, Nankai University, Tianjin, China | ³School of International Business, Tianjin Foreign Studies University, Tianjin, China**Correspondence:** Guang Yang (yangg_econ@163.com)**Received:** 11 December 2025 | **Revised:** 11 December 2025 | **Accepted:** 21 January 2026**Keywords:** cross-industry input | degree of robotisation | export performance | foreign robot usage | trade costs

ABSTRACT

This article utilises customs data from 2010 to 2016 to measure export values at the ‘city-industry’ level, focusing on the impact of industrial robot adoption in developed market-oriented countries on China’s export performance. A theoretical model is constructed to explore how the use of industrial robots abroad affects China’s exports. Additionally, a penetration index of foreign industrial robot usage and an instrumental variable are developed. The main findings are as follows: First, the use of robots in developed market-oriented countries has a negative impact on China’s exports, while the impact of robot adoption in competitive countries with similar endowments to China is not significant. Second, the negative effects from robot usage in developed countries are intensified in industries with a high cross-industry input ratio, low international trade costs, and a high degree of robotisation. Third, robot adoption in developed countries also poses certain obstacles to China’s efforts in exploring new export markets. This article has significant implications for formulating policies aimed at ‘stabilizing foreign trade’ and provides insights into the new dynamics of global industrial chain restructuring.

1 | Introduction

With the advancement of digital technology, new production technologies represented by industrial robots are rapidly maturing. Industrial robots are defined as ‘automated machinery that can be programmed along three or more axes, is automatically controlled, and can be either fixed or mobile for various applications’. Compared with traditional equipment like CNC machines, industrial robots significantly enhance mechanical autonomy. They can efficiently and effectively complete production tasks without requiring additional workers for operation. At present, industrial robots are increasingly substituting for workers in simple and routine tasks, and more complex tasks are also becoming automated. The cost-effectiveness of industrial robots is improving by approximately 8% per year. For instance, the average hourly wage of a skilled welder in developed countries is around \$25, while the average hourly operating cost of a

spot-welding robot is only about \$8, and it is expected to drop to \$2 by 2030 (Sirkin et al. 2015).

The widespread adoption of industrial robots will lead to a substantial replacement of labour in production (Frey and Osborne 2017), triggering adjustments in relative productivity and relative factor costs between countries. On the one hand, countries with factor endowments that are more compatible with industrial robot technology can fully capitalise on the economic benefits brought by industrial robots, leading to rapid gains in production efficiency and widening the gap with other nations (Acemoglu and Zilibotti 2001). On the other hand, as the costs of industrial robots continue to decline, the substitution of robots for labour diminishes the labour cost advantage, challenging the traditional Heckscher-Ohlin trade theory. These shifts will impact the global industrial layout of multinational enterprises, leading to changes

in trade flows (Stemmler 2019; Faber 2020; Krenz et al. 2021; Artuc et al. 2023). A typical example is the case of Adidas, which established automated factories primarily utilising industrial robots in Ansbach, Germany and Atlanta, USA. These facilities produce 500,000 pairs of shoes annually while employing only 160 people locally. In contrast, factories with the same output in Southeast Asian countries like Indonesia require over 1000 workers. The establishment of smart factories allows for a portion of previously imported apparel to be produced domestically.

Currently, some scholars have focused on the ‘re-offshoring’ effect brought about by the use of industrial robots, but research specifically targeting China remains limited. On the one hand, China actively attracted foreign investment for processing trade during the reform and opening-up period, and the shift towards ‘domestic production’ in developed countries has a negative impact on China’s exports. On the other hand, China has become a major global economy deeply integrated into the global value chain (GVC), and the adoption of industrial robots in developed countries can generate positive upstream and downstream spillovers, positively influencing China’s exports (Artuc et al. 2023). At present, China is at a critical stage of industrial transformation and upgrading, and ‘stabilising exports’ remains a crucial economic development goal. Therefore, to clarify the specific impact of industrial robot adoption in developed countries on the Chinese economy, this article constructs a multi-sector Ricardian model with task assignments for theoretical analysis, and following the methodology of Faber (2020), conducts an empirical study on the effect of industrial robot usage in developed countries on China’s export performance.¹

The most relevant literature to this article is Faber (2020), which examines the impact of US robot usage on the Mexican economy. Building on the framework of Acemoglu and Restrepo (2020), Faber incorporates international trade and constructs a general equilibrium model of the domestic labour market, including foreign robot penetration. The article finds that the adoption of robots in the United States significantly reduces Mexico’s employment growth rate and export volume. Additionally, Stemmler (2019) and Artuc et al. (2023) integrate the use of industrial robots with the Ricardian model, while Krenz et al. (2021) combine robot usage with the outsourcing model to investigate the trends in international trade under labour substitution by industrial robots. Ma et al. (2024) provides a quantitative study on the impact of global production automation on China’s trade and welfare. In addition, indirectly related literature includes studies on the factors influencing national industrial structures (Grossman and Rossi-Hansberg 2008; Ofori and Asongu 2024) and research on the economic impact of intelligent production technologies such as robotics (Acemoglu and Restrepo 2020; Oyewo et al. 2023).

The potential contributions of this article are as follows: First, this article derives the elasticity expression of bilateral trade changes caused by production cost variations due to robot adoption. This article constructs a multisector Ricardian model that incorporates robotic tasks and finds that under an international trade system, production cost changes in one country due to robot adoption influence the production costs of other countries through trade linkages, thereby leading to bilateral

trade changes. Among them, the ‘cross-country cost change elasticity’ weighted by bilateral trade shares is a key variable. If developed countries utilise robots to promote ‘production localisation’, the synchronisation channel of cross-border production cost changes weakens. The elasticity of the impact of China’s rising production costs on developed countries’ costs significantly decreases, while the elasticity of the impact of developed countries’ cost reductions on China’s costs significantly increases, ultimately leading to a decline in China’s exports to developed countries.

Second, this article constructs an indicator to measure the penetration of overseas robot adoption into the Chinese economy. In the research process, the study focuses on the impact of industrial robot usage in developed market economies on China. However, unlike Faber (2020), which examines the impact of US robot adoption on Mexico, China has established broader economic and trade linkages with countries worldwide. To ensure the accuracy of the empirical results, this article classifies overseas countries appropriately and controls for the impact of robot adoption in both China and competitor countries with similar endowments to China. Ultimately, this article selects countries that have relatively weak industrial division linkages with China but exhibit high levels of robot usage, constructing an instrumental variable suitable for the article.

Third, this article finds that the impact of robot adoption in developed countries on China’s exports is moderated by several key factors. Through an extended model analysis, it is revealed that the share of inter-industry intermediate input-output, international trade barriers, and the degree of substitutability of robots within industries significantly regulate this impact. Further empirical tests show that the negative effect of industrial robot adoption in developed market economies on China’s exports is particularly pronounced in industries characterised by a high proportion of inter-industry input, low trade costs, and a high degree of robotisation. For example, the export sector of the electronics, electrical appliances, and instrumentation manufacturing industry is particularly affected.

The remainder of this article is organised as follows: Section 2 presents the construction of the Ricardian model incorporating task assignments; Section 3 focuses on the empirical design, indicator construction, and data description; Section 4 provides an analysis of the empirical results; Section 5 covers the extension analysis; and Section 6 concludes with policy recommendations.

2 | Multisector Ricardian Model With Task Assignments

In the economic environment, there are N countries and J industries. Here, n and i represent individual countries, and j and k represent individual industries.

Labour and capital can flow freely across industries but not across countries. The labour supply in country n is L_n , the labour supply in industry j in country n is L_n^j , and the amount of labour employed by a representative firm in industry j in country n is l_n^j . Additionally, robots are purchased with domestically allocated capital. The robot stock in country n is H_n , the robot stock in

industry j in country n is H_n^j , and the number of robots used by a representative firm in industry j in country n is h_n^j .

2.1 | Production Tasks and Composite Factors

Following Acemoglu and Restrepo (2018), this article assumes that intermediate goods firms must first combine robots and labour into a composite factor to complete various tasks in production. Suppose the economic environment is perfectly competitive, and the set of production tasks is standardised to a unit interval, with each ‘point’ task denoted as $o \in [\varpi_n^j - 1, \varpi_n^j]$. An increase in ϖ_n^j implies an improvement in the overall quality of production tasks in industry j in country n , potentially reflecting the emergence of new types of activities. Now, suppose that the ‘point’ tasks in $[\varpi_n^j - 1, \varpi_n^j]$ are ordered from low to high complexity, with $\bar{O}_n^j \in [\varpi_n^j - 1, \varpi_n^j]$ representing a critical value indicating the most complex task that can be performed by robots in industry j in country n . As the value of $\bar{O}_n^j$ increases, it indicates that robots can replace more labour tasks.

Low-complexity tasks $\varpi_n^j - 1 \leq o_n^j \leq \bar{O}_n^j$ can be completed by robots as a substitute for labour:

$$t_n^j(o) = \bar{B}(\zeta^j) \left[(\iota^j)^{\frac{1}{\zeta^j}} \Theta^j(o)^{\frac{\zeta^j-1}{\zeta^j}} + (1-\iota^j)^{\frac{1}{\zeta^j}} (h_n^j(o) + A_n^j(o)l_n^j(o))^{\frac{\zeta^j-1}{\zeta^j}} \right]^{\frac{\zeta^j}{\zeta^j-1}}$$

In this equation, $\bar{B}(\zeta^j)$ is a simplifying constant, $\Theta^j(o)$ represents the specific factor required to complete ‘point’ task, $\zeta^j \in (0, \infty)$ is the elasticity of substitution between the specific factor and labour and robots, and $\iota^j \in (0, 1)$ is the share parameter of the specific factor.² ζ^j and ι^j vary only across industries. $A_n^j(o)$ is the labour productivity of industry j in country n ; robots can complete tasks of different complexity with similar productivity, whereas labour productivity differs more significantly across tasks of varying complexity. It is assumed that $A_n^j(o)$ strictly increases with the complexity of ‘point’ task o . The term $k_n^j(o) + A_n^j(o)l_n^j(o)$ indicates a complete substitutability between robots and labour, where firms choose to use one factor based on cost minimisation and technological constraints. Additionally, $l_n^j(o)$ and $h_n^j(o)$ represent the labour and robot inputs required by firms to complete ‘point’ task o .

High-complexity tasks $\bar{O}_n^j < o \leq \varpi_n^j$ can only be completed by labour:

$$t_n^j(o) = \bar{B}(\zeta^j) \left[(\iota^j)^{\frac{1}{\zeta^j}} \Theta^j(o)^{\frac{\zeta^j-1}{\zeta^j}} + (1-\iota^j)^{\frac{1}{\zeta^j}} (A_n^j(o)l_n^j(o))^{\frac{\zeta^j-1}{\zeta^j}} \right]^{\frac{\zeta^j}{\zeta^j-1}}$$

In the profit maximisation problem, the cost (or price) $p_n^j(o)$ for a representative firm to complete ‘point’ task o can be derived as a piecewise function: when $o \in [\varpi_n^j - 1, \bar{O}_n^j]$:

$$p_n^j(o) = \left[\iota^j (\vartheta_n^j)^{1-\zeta^j} + (1-\iota^j) \min \left\{ r_n, \frac{w_n}{A_n^j(o)} \right\} \right]^{\frac{1}{1-\zeta^j}}$$

when $o \in [\bar{O}_n^j, \varpi_n^j]$:

$$p_n^j(o) = \left[\iota^j (\vartheta_n^j)^{1-\zeta^j} + (1-\iota^j) \left(\frac{w_n}{A_n^j(o)} \right)^{1-\zeta^j} \right]^{\frac{1}{1-\zeta^j}}$$

At this point, it is necessary to compare the costs of robots and labour. Assume there exists $\tilde{O}_n^j$ such that $r_n = \frac{w_n}{A_n^j(\tilde{O}_n^j)}$. This yields the critical threshold between robots and labour, $O_n^{j*} = \min \{ \bar{O}_n^j, \tilde{O}_n^j \}$, leading to:

If $O_n^{j*} = \min \{ \bar{O}_n^j, \tilde{O}_n^j \} = \tilde{O}_n^j < \bar{O}_n^j$, it is referred to as ‘factor price-driven automation’.

If $O_n^{j*} = \min \{ \bar{O}_n^j, \tilde{O}_n^j \} = \bar{O}_n^j < \tilde{O}_n^j$, it is referred to as ‘technology-driven automation’.

When $o \in [\varpi_n^j - 1, O_n^{j*}]$:

$$p_n^j(o) = \left[\iota^j (\vartheta_n^j)^{1-\zeta^j} + (1-\iota^j) (r_n)^{1-\zeta^j} \right]^{\frac{1}{1-\zeta^j}}$$

when $o \in [O_n^{j*}, \varpi_n^j]$:

$$p_n^j(o) = \left[\iota^j (\vartheta_n^j)^{1-\zeta^j} + (1-\iota^j) \left(\frac{w_n}{A_n^j(o)} \right)^{1-\zeta^j} \right]^{\frac{1}{1-\zeta^j}}$$

A representative firm in industry j in country n combines ‘point’ tasks $t_n^j(o)$ into a composite factor y_n^j in a CES (constant elasticity of substitution) form:

$$y_n^j = \tilde{B} \left[\int_{\varpi_n^j-1}^{\varpi_n^j} t_n^j(o)^{\frac{\eta^j-1}{\eta^j}} do \right]^{\frac{\eta^j}{\eta^j-1}}$$

Here, η^j is the elasticity of substitution for ‘point’ tasks in the composite process, and η^j varies only across industries. A representative firm in industry j in country n utilises ‘point’ tasks $t_n^j(o)$ to construct the composite factor y_n^j , resulting in the demand function for each ‘point’ task as follows:

$$t_n^j(o) = \tilde{B}^{\eta^j-1} p_n^j(o)^{-\eta^j} y_n^j$$

Using the above equation and based on the assumptions $\iota^j \rightarrow 0$ or $\zeta^j = 1$ from Acemoglu and Restrepo (2018),³ the demand functions for robots and labour to complete ‘point’ tasks are derived as piecewise functions: when $o \in [\varpi_n^j - 1, O_n^{j*}]$, $h_n^j(o) = (1-\iota^j) B^{\eta^j-1} y_n^j (r_n)^{-\eta^j}$; $o \in [O_n^{j*}, \varpi_n^j]$, $h_n^j(o) = 0$; $o \in [\varpi_n^j - 1, O_n^{j*}]$, $l_n^j(o) = (1-\iota^j) B^{\eta^j-1} y_n^j \frac{1}{A_n^j(o)} \left(\frac{w_n}{A_n^j(o)} \right)^{-\eta^j}$ and $o \in [O_n^{j*}, \varpi_n^j]$, $l_n^j(o) = 0$.

Here, $\hat{\eta}^j = \eta^j(1 - \iota^j)$, $B = \tilde{B}^{\frac{\hat{\eta}^j-1}{\hat{\eta}^j}}$; under the assumption $\iota^j \rightarrow 0$, we have $\hat{\eta}^j = \eta^j$.

Given the demand functions for robots and labour in ‘point’ tasks, the robot and labour inputs of a representative firm in the construction of the composite factor are as follows:

$$h_n^j = \int_{\varpi_n^*-1}^{O_n^{j*}} (1 - \iota^j) B^{\hat{\eta}^j-1} y_n^j(r_n)^{-\hat{\eta}^j} do \\ = (1 - \iota^j) B^{\hat{\eta}^j-1} y_n^j(r_n)^{-\hat{\eta}^j} (O_n^{j*} + 1 - \varpi_n^j)$$

$$l_n^j = \int_{O_n^{j*}}^{\varpi_n^j} (1 - \iota^j) B^{\hat{\eta}^j-1} y_n^j \frac{1}{A_n^j(o)} \left(\frac{w_n}{A_n^j(o)} \right)^{-\hat{\eta}^j} do = (1 - \iota^j) B^{\hat{\eta}^j-1} y_n^j \int_{O_n^{j*}}^{\varpi_n^j} \frac{1}{A_n^j(o)} \left(\frac{w_n}{A_n^j(o)} \right)^{-\hat{\eta}^j} do$$

From the conditions of perfect competition and market clearing, we have $r_n h_n^j + w_n l_n^j = (1 - \iota^j) y_n^j$. The ideal price index is then:

$$(r_n)^{1-\hat{\eta}^j} (O_n^{j*} + 1 - \varpi_n^j) + \int_{O_n^{j*}}^{\varpi_n^j} \left(\frac{w_n}{A_n^j(o)} \right)^{1-\hat{\eta}^j} do = B^{1-\hat{\eta}^j}$$

In summary, we can establish a direct functional relationship between the composite factor and the inputs of robots and labour, as follows:

$$y_n^j = \left[(\phi_n^j)^{\frac{1}{\hat{\eta}^j}} (h_n^j)^{\frac{\hat{\eta}^j-1}{\hat{\eta}^j}} + (1 - \phi_n^j)^{\frac{1}{\hat{\eta}^j}} (l_n^j)^{\frac{\hat{\eta}^j-1}{\hat{\eta}^j}} \right]^{\frac{\hat{\eta}^j}{\hat{\eta}^j-1}}$$

Here, $\phi_n^j = O_n^{j*} - \varpi_n^j + 1$ and $1 - \phi_n^j = \int_{O_n^{j*}}^{\varpi_n^j} A_n^j(o)^{\hat{\eta}^j-1} do$, where y_n^j represents the quantity of the composite factor for a representative firm in industry j in country n ; h_n^j and l_n^j denote the robot capital and labour inputs, respectively, required by a representative firm in industry j in country n to produce the composite

factor. Additionally, the parameter $\hat{\eta}^j \in [0, \infty)$ represents the elasticity of substitution between robots and traditional labour, and $\phi_n^j \in (0, 1)$ is the share parameter. In the studies by Eaton and Kortum (2002) and Caliendo and Parro (2015), for simplification, it is assumed that firms only use labour for production. In this article, to incorporate robots, the capital used by firms to purchase robots is treated separately and combined with labour to form a composite factor. Using these conditions, the price of the composite factor (ρ_n^j) can be solved as follows:

$$\rho_n^j = \left[(\phi_n^j (r_n)^{1-\hat{\eta}^j} + (1 - \phi_n^j) (w_n)^{1-\hat{\eta}^j}) \right]^{\frac{1}{1-\hat{\eta}^j}}$$

2.2 | Intermediate Goods and Trade Cost

In the production process, intermediate goods firms need to input raw materials and then either hire labour or purchase robots to complete production tasks. In industry j , there exists a set of intermediate goods ω^j , where $\omega^j \in [0, 1]$. Each intermediate goods category can be viewed as a ‘point’ product along the product interval, with the production technology given by:

$$q_n^j(\omega^j) = z_n^j(\omega^j) [y_n^j(\omega^j)]^{\gamma_n^j} \prod_{k=1}^J [m_n^{kj}(\omega^j)]^{\gamma_n^{kj}}$$

Here, $q_n^j(\omega^j)$ represents the output of a representative firm in industry j in country n for ‘point’ intermediate good ω^j ; $z_n^j(\omega^j)$ is the production efficiency of a representative firm in industry j in country n for ‘point’ intermediate good ω^j ; $y_n^j(\omega^j)$ represents the input quantity of the composite factor; and $m_n^{kj}(\omega^j)$ represents the quantity of composite intermediate goods from industry k input into industry j . γ_n^j denotes the income share of the composite factor, and γ_n^{kj} represents the income share of composite intermediate goods from industry k , and these satisfy $\gamma_n^j + \sum_{k=1}^J \gamma_n^{kj} = 1$.

Following Caliendo and Parro (2015), under constant returns to scale and perfect competition, the price of intermediate goods p_n^j and the unit cost of the input bundle c_n^j are given by:

$$p_n^j = \frac{Y_n^j}{z_n^j} (\rho_n^j)^{\gamma_n^j} \prod_{k=1}^J (P_n^k)^{\gamma_n^{kj}}$$

$$c_n^j = Y_n^j (\rho_n^j)^{\gamma_n^j} \prod_{k=1}^J (P_n^k)^{\gamma_n^{kj}}$$

Substituting the expression for ρ_n^j into c_n^j yields:

$$c_n^j = Y_n^j \left[(O_n^{j*} - \varpi_n^j + 1) (r_n)^{1-\hat{\eta}^j} + \int_{O_n^{j*}}^{\varpi_n^j} A_n^j(o)^{\hat{\eta}^j-1} do \right] (w_n)^{1-\hat{\eta}^j} \prod_{k=1}^J (P_n^k)^{\gamma_n^{kj}} \quad (1)$$

We assume that ‘point’ intermediate goods in each industry can be traded internationally. When industries combine composite intermediate goods, the ‘point’ intermediate goods used may be produced domestically or abroad. The process of transporting ‘point’ intermediate goods from country i to country n incurs two types of costs: an iceberg cost $d_{in}^j \geq 1$ and an ad valorem tariff τ_{in}^j , so the total trade cost is $\kappa_{in}^j = (1 + \tau_{in}^j) d_{in}^j = \tilde{\tau}_{in}^j d_{in}^j$, with no arbitrage from re-exports.

The landed price of intermediate goods produced by industry j in country i and exported to country n is determined by selecting the lowest-cost option from all countries:

$$p_n^j = \min_i \left\{ p_{in}^j = \frac{c_i^j k_{in}^j}{z_i^j} \right\}$$

Following Eaton and Kortum (2002) and Caliendo and Parro (2015), and leveraging the properties of the Frechet distribution function, this article derives the price of composite intermediate goods P_n^j , based on international trade in 'point' intermediate goods, as follows:

$$P_n^j = \xi^j \left[\sum_{i=1}^N \mu_i^j (c_i^j k_{in}^j)^{-\theta^j} \right]^{-\frac{1}{\theta^j}}$$

Here, $\xi^j = \Gamma\left(\frac{1-\sigma^j+\theta^j}{\theta^j}\right)^{\frac{1}{1-\sigma^j}}$, μ_n^j is the location parameter of the Frechet distribution, and θ^j is the shape parameter of the Frechet distribution. This article also derives the share of inputs from country i in industry j as a proportion of country n 's total:

$$\pi_{in}^j = \frac{X_{in}^j}{X_n^j} = \frac{\mu_i^j (c_i^j k_{in}^j)^{-\theta^j}}{\sum_{i=1}^N \mu_i^j (c_i^j k_{in}^j)^{-\theta^j}} \quad (2)$$

2.3 | The Impact of Robot Usage on Costs

In each industry j within each country n worldwide, some routine production tasks are performed by robots. Assuming uniform technological frontiers for robots across countries due to capital goods trade and talent exchange, but differing endowments of labour and capital between countries, O_n^{j*} will vary at the national level. The extent to which robots replace labour differs across countries, resulting in different composite factor inputs γ_n^j for producing intermediate goods, which in turn leads to varying unit input bundle costs c_n^j . Consequently, this affects trade flows π_{in}^j and trade volumes X_{in}^j . This section examines the impact of varying levels of robot usage on intermediate goods costs in industry j in country n .

This article focuses on the impact of changes in O_n^{j*} on production costs. Under other constant conditions, in Equation (1), we take the partial derivative of the unit input bundle cost c_n^j for intermediate goods in industry j of country n with respect to the domestic automation level O_n^{j*} , yielding:

$$\frac{\partial c_n^j}{\partial O_n^{j*}} = \underbrace{\frac{c_n^j \gamma_n^j f^{\frac{\theta^j-1}{\theta^j}} \left[(r_n)^{1-\tilde{\eta}^j} - \left(\frac{w_n}{A_n^j(O_n^{j*})} \right)^{1-\tilde{\eta}^j} \right]}{1-\tilde{\eta}^j}}_{\text{Factor Adjustment Effect}} + \underbrace{c_n^j \sum_{k=1}^J \gamma_n^{kj} (P_n^k)^{\theta^k} \left[\sum_{i=1}^N \mu_i^k (c_i^k)^{-\theta^k-1} (\kappa_{in}^k)^{-\theta^k} \left(\frac{\partial c_i^k}{\partial O_n^{j*}} \right) \right]}_{\text{Industry Linkage Effect}} \quad (3)$$

2.3.1 | Factor Adjustment Effect

The first term on the right side of the equation is referred to as the 'factor adjustment effect'. This effect indicates that when O_n^{j*} in industry j of country n changes, the share of factor inputs is reallocated as some tasks are completed by robots, which

subsequently induces a change in production cost c_n^j . When $O_n^{j*} = \min \left\{ \bar{O}_n^j, \tilde{O}_n^j \right\}$ and $r_n = \frac{w_n}{A_n^j(\tilde{O}_n^j)} \Rightarrow A_n^j(\tilde{O}_n^j) = \frac{w_n}{r_n}$, we obtain:

1. If $O_n^{j*} = \min \left\{ \bar{O}_n^j, \tilde{O}_n^j \right\} = \tilde{O}_n^j < \bar{O}_n^j$, meaning that the increase in robot usage is driven by changes in factor prices, such as an increase in labour costs w_n or a decrease in capital rental rate r_n , then $(r_n)^{1-\tilde{\eta}^j} - \left(\frac{w_n}{A_n^j(O_n^{j*})} \right)^{1-\tilde{\eta}^j} = 0$.
2. If $O_n^{j*} = \min \left\{ \bar{O}_n^j, \tilde{O}_n^j \right\} = \bar{O}_n^j < \tilde{O}_n^j$, meaning that the increase in robot usage is driven by advancements in robot technology, then $(r_n)^{1-\tilde{\eta}^j} - \left(\frac{w_n}{A_n^j(O_n^{j*})} \right)^{1-\tilde{\eta}^j} < 0$.

2.3.2 | Industry Linkage Effect

The second term on the right side of the equation represents the 'industry linkage effect' induced by O_n^{j*} . When there is no cross-industry input or international trade, this term equals zero. The sign of this effect depends on $\frac{\partial c_i^k}{\partial O_n^{j*}}$, and the analytical approach is as follows: First, in the absence of cross-industry input and international trade, the impact of domestic robot usage in the same industry on domestic production costs can be represented as $\frac{\partial c_n^j}{\partial O_n^{j*}}$. Second, when cross-industry input and international trade are present, due to the industry linkage effect, changes in production costs within the same industry in the same country will induce corresponding changes in production costs in other industries and other countries, that is, $\frac{\partial c_i^k}{\partial O_n^{j*}}$, which then translates to $\frac{\partial c_i^k}{\partial O_n^{j*}}$ and can be written as: $\sum_{k=1}^J \gamma_n^{kj} (P_n^k)^{\theta^k} \left[\sum_{i=1}^N \mu_i^k (c_i^k)^{-\theta^k-1} (\kappa_{in}^k)^{-\theta^k} \frac{\partial c_i^k}{\partial O_n^{j*}} \right]$. Finally, $\frac{\partial c_i^k}{\partial O_n^{j*}}$ once again influences $\frac{\partial c_n^j}{\partial O_n^{j*}}$ through cross-industry and cross-country linkage effects. This article classifies the analysis into three cases:

1. If $O_n^{j*} = \min \left\{ \bar{O}_n^j, \tilde{O}_n^j \right\} = \bar{O}_n^j < \tilde{O}_n^j$, this means that the increase in robot usage is entirely driven by advancements in robot technology. In the absence of international trade and cross-industry input, the production costs in industry i in country n decrease. When international trade and cross-industry input are introduced, the cost reduction effect

spills over. Given that the coefficient of each additive term, $\gamma_n^{kj} \mu_i^k (\kappa_{in}^k)^{-\theta^k} (c_i^k)^{-\theta^k-1} (\xi^k)^{-\theta^k} (P_n^k)^{\theta^k} > 0$, it ultimately follows that $\frac{\partial c_n^j}{\partial O_n^{j*}} < 0$. Representative countries include the United States, Western Europe, and Japan, where labour costs are higher than capital rental rates. If breakthroughs

in robot technology are achieved, they can significantly reduce the production costs of intermediate goods.

2. If $O_n^{j*} = \min \{\bar{O}_n^j, \tilde{O}_n^j\} = \tilde{O}_n^j < \bar{O}_n^j$, this indicates that the increase in robot usage is driven by changes in factor prices. If the change in factor prices is due to an increase in labour costs w_n , this implies that in the absence of international trade and cross-industry input, the production costs in industry j in country n will rise. When international trade and cross-industry input are introduced, the cost increase effect spills over. It ultimately follows that $\frac{\partial c_n^j}{\partial O_n^{j*}} \geq 0$. Representative countries include labour-intensive developing countries like China; if rising labour costs force the adoption of robots, it could weaken China's competitiveness in the international market.
3. If $O_n^{j*} = \min \{\bar{O}_n^j, \tilde{O}_n^j\} = \bar{O}_n^j < \tilde{O}_n^j$, this indicates that the increase in robot usage is driven by changes in factor prices. If the change in factor prices is due to a decrease in the capital rental rate r_n , this implies that in the absence of international trade and cross-industry input, the production costs in industry j in country n will decrease. When international trade and cross-industry input are introduced, the cost reduction effect spills over. It ultimately follows that $\frac{\partial c_n^j}{\partial O_n^{j*}} \geq 0$.

This article also takes the partial derivative of the unit cost of the input bundle c_i^k for intermediate goods in industry k in country i with respect to the automation level O_n^{j*} in industry j in country n , yielding: $\frac{\partial c_i^k}{\partial O_n^{j*}} = c_i^k \left(\sum_{h=1}^J \gamma_i^{hk} (P_i^h)^{\theta^h} \left[\sum_{m=1}^N \mu_m^h (c_m^h)^{-\theta^h-1} (\kappa_{mi}^h)^{-\theta^h} \frac{\partial c_m^h}{\partial O_n^{j*}} \right] \right)$. This reveals that only cross-country and cross-industry production linkage effects are present. Furthermore, the impact of $\frac{\partial c_m^h}{\partial O_n^{j*}}$ aligns with the previous assessment method and depends on the constraint form of the robot applicability boundary O_n^{j*} .

In summary, in the absence of cross-industry input and international trade, the production cost c_n^j in industry j in country i depends on the mode of change in domestic robot usage O_n^{j*} . The 'factor cost-driven' and 'technology-driven' modes determine different directions for changes in c_n^j . When cross-industry input and international trade are present, changes in c_n^j will, through linkage effects, cause production costs c_i^k in other industries and other countries to change in the same direction, that is, $\frac{\partial c_i^k}{\partial c_n^j} > 0$. Changes in c_i^k imply an adjustment in the global cost structure, which in turn influences the effect of robots on costs, $\frac{\partial c_n^j}{\partial O_n^{j*}}$. Currently, the primary reason for industrial robot adoption in both developed market-oriented countries and China is the rise in labour costs. See Table 1 for details.

TABLE 1 | Impact of robot usage on costs.

Constraint form	Change of O_n^{j*}	Factor adjustment effect	Impact on other c_i^k	Industry linkage effect
$O_n^{j*} = \min \{\bar{O}_n^j, \tilde{O}_n^j\} = \tilde{O}_n^j$	When $O_n^{j*} \uparrow$	= 0	$r_n \downarrow, \frac{\partial c_i^k}{\partial O_n^{j*}} \leq 0$	< 0
	When $O_n^{j*} \uparrow$	= 0	$w_n \uparrow, \frac{\partial c_i^k}{\partial O_n^{j*}} \geq 0$	> 0
$O_n^{j*} = \min \{\bar{O}_n^j, \tilde{O}_n^j\} = \bar{O}_n^j$	When $O_n^{j*} \uparrow$	< 0	Technological $\uparrow, \frac{\partial c_i^k}{\partial O_n^{j*}} \leq 0$	< 0

2.4 | Impact of Cost Changes on Trade

As long as there is labour cheaper than robots somewhere in the world, multinational firms will employ labour for production. The fundamental reason for the adoption of industrial robots in the home countries of multinationals is that overseas labour is no longer inexpensive. Based on this, this article constructs a two-stage analytical framework. In the first stage, rising labour costs in emerging countries like China lead to increased production costs. In the second stage, major multinational home countries in regions like Europe and the United States adopt industrial robots to mitigate the risks associated with rising overseas labour costs.

2.4.1 | In the First Stage

For product-exporting country c_i^j , based on Equation (2), taking the derivative of the import share of country n from country i in industry j with respect to the unit production cost c_i^j of intermediate goods in industry j in country i yields:

$$\frac{\partial \pi_{in}^j}{\partial c_i^j} = \frac{\theta^j \mu_i^j (c_i^j \kappa_{in}^j)^{-\theta^j}}{c_i^j} \frac{\sum_{m \neq i}^N \mu_m^j (c_m^j \kappa_{mm}^j)^{-\theta^j}}{\left[\sum_{i=1}^N \mu_i^j (c_i^j \kappa_{in}^j)^{-\theta^j} \right]^2} (\Delta_{im}^j - 1) \quad (4)$$

Here, $\Delta_{im}^j = \frac{c_i^j}{c_m^j} \frac{\partial c_m^j}{\partial c_i^j}$ is the cost change elasticity, indicating the extent to which a change in the intermediate input cost c_i^j in industry j of country i affects the intermediate input cost c_m^j in country m . The subscript m represents a country, just like i and n . When $\Delta_{im}^j < 1$, the change in unit cost c_i^j in industry j of country i has a minimal effect on the costs of other countries c_m^j . In this case, if c_i^j decreases, country n will increase its imports from country i in industry j . When $\Delta_{im}^j > 1$, the change in unit cost c_i^j in industry j of country i significantly affects the costs c_m^j in other countries. In this scenario, if c_i^j decreases, country n will reduce its imports from country i in industry j .

For China, we assume a constraint form of $O_n^{j*} = \min \{\bar{O}_n^j, \tilde{O}_n^j\} = \tilde{O}_n^j$, meaning that the primary purpose of robot adoption is to address the rising domestic labour costs. This, in turn, implies $\frac{\partial c_n^j}{\partial O_n^{j*}} > 0$ and $\frac{\partial c_i^k}{\partial O_n^{j*}} > 0$. Furthermore, when domestic intermediate goods production costs increase, they raise foreign intermediate goods production costs through international trade and cross-industry input linkages. When cost elasticity $\Delta_{im}^j < 1$, the trade volume (or trade occurrence probability) π_{in}^j decreases; when cost elasticity $\Delta_{im}^j > 1$, the trade volume increases

instead of decreasing. By rearranging the equation (4), we obtain the elasticity of Chinese cost on Sino-US trade π_{in}^j :

$$\frac{c_i^j}{\pi_{in}^j} \frac{\partial \pi_{in}^j}{\partial c_i^j} = \theta^j \left[\sum_{\substack{m \neq i \\ m \neq n}}^N \pi_{mn}^j (\Delta_{im}^j - 1) + \pi_{nn}^j (\Delta_{in}^j - 1) \right] \quad (5)$$

Combining Equation (5) with the actual situation, the rise in c_i^j due to increased labour costs in China i prompts multinational parent countries n , such as those in Europe and the United States, to substitute labour with robots to mitigate the impact of rising c_i^j . In this case, the cost elasticity Δ_{in}^j approaches zero, and $\Delta_{in}^j - 1$ approaches -1 . In the case of large countries, the self-sufficiency share is typically much greater than the import share from third countries, $\pi_{nn}^j > \sum_{m \neq i}^N \pi_{mn}^j$, making $\frac{c_i^j}{\pi_{in}^j} \frac{\partial \pi_{in}^j}{\partial c_i^j} \leq 0$ hold true.

2.4.2 | In the Second Stage

For the product-exporting country c_n^j , based on Equation (2), taking the derivative of the import share of country n from country i in industry j with respect to the unit production cost c_n^j of intermediate goods in industry j in country n yields:

$$\frac{\partial \pi_{in}^j}{\partial c_n^j} = \frac{\theta^j \mu_i^j (c_i^j \kappa_{in}^j)^{-\theta^j} \sum_{m \neq i}^N \mu_m^j (c_m^j \kappa_{mn}^j)^{-\theta^j}}{c_n^j \left[\sum_{i=1}^N \mu_i^j (c_i^j \kappa_{in}^j)^{-\theta^j} \right]^2} (\Delta_{nm}^j - \Delta_{ni}^j) \quad (6)$$

Thus, when the cost in industry j of country n increases (or decreases), it causes costs in other countries to increase (or decrease) as well, with $\Delta_{nm}^j \geq 0$. If the increase (or decrease) in the costs of other countries m is greater than the increase (or decrease) in the costs of country i , so that $\Delta_{nm}^j - \Delta_{ni}^j \geq 0$, then the trade volume between country n and country i will increase.

For example, if the use of industrial robots causes a 10% decrease in c_n^j for the United States (n), this reduction in c_n^j may cause a 4% decrease in c_i^j for China (i). If it also causes a 6% decrease in c_{m1}^j for Vietnam ($m1$), then Vietnam is considered to have an advantage over China. Conversely, if it results in only a 3% decrease in c_{m2}^j for the Philippines ($m2$), then the Philippines is considered to have a disadvantage relative to China. By weighting the advantages and disadvantages of all countries in the world (including the United States itself) except China, if the weighted cost reduction overall exceeds 4%, China's export volume to the United States will decline, and vice versa. Similarly, after modifying Equation (6), we obtain the elasticity of US costs with respect to Sino-US trade π_{in}^j as follows:

$$\frac{c_n^j}{\pi_{in}^j} \frac{\partial \pi_{in}^j}{\partial c_n^j} = \theta^j \left[\sum_{\substack{m \neq i \\ m \neq n}}^N \pi_{mn}^j (\Delta_{nm}^j - \Delta_{ni}^j) + \pi_{nn}^j (1 - \Delta_{ni}^j) \right] \quad (7)$$

Combining Equation (7) with the actual situation, due to rising labour costs in China i , the production cost c_i^j increases. In response,

multinational parent countries n , such as those in Europe and the United States, substitute labour with robots to mitigate the impact of the c_i^j increase. In this case, the cost elasticity Δ_{ni}^j approaches positive infinity, and $1 - \Delta_{ni}^j$ approaches negative infinity. Under the large-country assumption, the self-sufficiency share is typically greater than the import share from third countries, $\pi_{nn}^j > \sum_{m \neq i}^N \pi_{mn}^j$, making $\frac{c_i^j}{\pi_{in}^j} \frac{\partial \pi_{in}^j}{\partial c_i^j} \leq 0$ hold true.

$m \neq n$

Based on the above analysis, we propose

Proposition 1. *The use of robots leads to relative cost changes between countries, affecting the scale of China's exports to developed market-oriented countries. The specific impact depends on the elasticity of cost changes between countries.*

2.5 | Extension Analysis

2.5.1 | Moderating Effect of γ_n^{kj} and κ_{in}^k

We find that, within the input-output effects across countries and industries, the magnitude of the impact of changes in production costs c_i^k in other countries and industries on $\frac{\partial c_n^j}{\partial c_i^k}$ is determined by a weighted sum of $N \times J$ terms $\frac{\partial c_i^k}{\partial c_n^j}$ according to input-output relationships. The weights can be represented as $\Psi = \gamma_n^{kj} \mu_i^k (c_i^k)^{-\theta^k-1} (\kappa_{in}^k)^{-\theta^k} (\xi^k)^{-\theta^k} (P_n^k)^{\theta^k}$.

First, this weight is influenced by γ_n^{kj} , which represents the share of inputs from industry k in total raw material inputs used by industry j in country n when producing intermediate goods. Taking the partial derivative of the weight with respect to γ_n^{kj} yields:

$$\frac{\partial \Psi}{\partial \gamma_n^{kj}} = \mu_i^k (\kappa_{in}^k)^{-\theta^k} (c_i^k)^{-\theta^k-1} (\xi^k)^{-\theta^k} (P_n^k)^{\theta^k} > 0$$

where c_i^k and P_n^k do not contain γ_n^{kj} in their expressions. As the derivative is greater than zero, the greater the proportion γ_n^{kj} of inputs from industry k used by industry j in country n during production, the stronger the input-output linkage effect, further amplifying the impact of robots on the production cost of intermediate goods.

Second, this weight is also influenced by κ_{in}^k , which represents the trade cost for products from industry k exported from country i to country n . Taking the partial derivative of the weight with respect to κ_{in}^k yields:

$$\frac{\partial \Psi}{\partial \kappa_{in}^k} = \gamma_n^{kj} \mu_i^k (c_i^k \kappa_{in}^k)^{-\theta^k-1} \left(\frac{P_n^k}{\xi^k} \right)^{\theta^k} \theta^k \left[-1 + \frac{\mu_i^k (\kappa_{in}^k c_i^k)^{-\theta^k}}{\left(\frac{P_n^k}{\xi^k} \right)^{-\theta^k}} \right]$$

$$\begin{aligned} P_n^j &= \xi^j \left[\sum_{i=1}^N \mu_i^j (c_i^j \kappa_{in}^j)^{-\theta^j} \right]^{-\frac{1}{\theta^j}} \Rightarrow (P_n^j)^{-\theta^j} \\ &= \sum_{i=1}^N \mu_i^j (c_i^j \kappa_{in}^j)^{-\theta^j} \Rightarrow \frac{\mu_i^j (c_i^j \kappa_{in}^j)^{-\theta^j}}{\sum_{i=1}^N \mu_i^j (c_i^j \kappa_{in}^j)^{-\theta^j}} < 1 \end{aligned}$$

Therefore, $\frac{\partial \Psi}{\partial \kappa_{in}^j} < 0$. This implies that the higher the trade cost κ_{in}^j for intermediate goods traded from country i to country n in the production process of industry j in country n , the weaker the input–output linkage effect, thereby diminishing the impact of robots on the production cost of intermediate goods.

Based on this, we propose

Proposition 2. *The higher the dependency of industry j on intermediate goods from other industries, the stronger the impact of robot usage in industry j on the production cost of intermediate goods in that industry. Similarly, the lower the trade costs between country i and other countries, the stronger the impact of robot usage in country i on the production cost of intermediate goods in that country.*

2.5.2 | Heterogeneity in Industry Robotisability

This article assumes that robots and labour are combined to form a composite factor, which is then used to complete production tasks. Through derivation, the aggregation process of the composite factor satisfies the following CES form:

$$y_n^j = \frac{B}{1-\rho} \left[\left(O_n^{j*} - \varpi_n^j + 1 \right)^{\frac{1}{\rho}} \left(h_n^j \right)^{\frac{\rho-1}{\rho}} + \left(\int \frac{\varpi_n^j}{O_n^{j*}} A_n^j(o)^{\rho-1} do \right)^{\frac{1}{\rho}} \left(\frac{p_n^j}{O_n^{j*}} \right)^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}}$$

when $r_n < \frac{w_n}{A_n^j(o)}$, firms will use robots. As a result, a proportion

of tasks equal to $O_n^{j*} - \varpi_n^j + 1$ can be substituted by robots, thereby achieving cost savings. The robotisability of industry j in country n is measured by $O_n^{j*} - \varpi_n^j + 1$. For ease of analysis, this article holds ϖ_n^j constant, so the larger O_n^{j*} , the more tasks in this industry can be completed by robots. When developed countries use robots in the production process of industry j , the cost and trade changes resulting from this transformation in production methods become more significant.

Based on this, we propose

Proposition 3. *There is heterogeneity in the degree of robotisability across industries. In industries with a higher degree of robotisability, the impact of robot usage on trade volume changes is more pronounced.*

3 | Empirical Design, Indicator Construction, and Data Description

In the relationship between developed market-oriented countries and China, the former is typically positioned upstream in the GVC, supplying core technologies and components while transferring simpler production processes to the latter. China, in turn, performs processing based on the provided inputs and exports the final consumer goods back to the developed countries. When developed market-oriented countries adopt industrial robots, the final processing stages can be carried out domestically at lower costs, leading to a decline in China's export volume to

these countries. This article focuses on the impact of robot adoption in market-oriented countries with close economic and trade ties to China, examining how it affects China's 'city-industry' level exports. The study period spans from 2010 to 2016, which avoids major shocks like the 2008 global financial crisis and the 2018 US–China trade war, ensuring a relatively stable trade environment conducive to this research. Additionally, this period saw rapid advancements in robotics technology, making the analysis timely. The article focuses specifically on the manufacturing sector, covering industries classified under categories 13–43 of the 2017 GB codes, which correspond to the ISIC.4 manufacturing classifications used in reports by the International Federation of Robotics (IFR).

3.1 | Classification of Developed Countries

To accurately address the research question, it is necessary to control for robot usage outside of developed countries. (1) China's domestic robot usage: rising labour costs in China are a significant factor driving the adoption of industrial robots in developed countries, leading to localised supply and a reduction in China's exports to these markets. However, the use of robots within China can mitigate rising domestic production costs, thereby weakening the negative impact. (2) Robot usage in competitive countries with similar endowments to China: in the global production network, other emerging countries with low-cost labour endowments have also established production ties with developed countries, competing with China for access to developed markets. When competitive countries adopt robots, their production costs may decrease, intensifying market competition and negatively affecting China's exports to developed countries.

If the empirical model only includes robot usage in developed countries, the results would be confounded by the combined effects of robot usage in developed countries, competitive countries, and China, making it difficult to assess the true impact.

Based on this, it is necessary to reconstruct the measurement method for robot usage, drawing on the frameworks of Acemoglu and Restrepo (2020), Faber (2020), and Gihleb et al. (2022). A critical step is the regional classification of market-oriented and competitive countries.

For developed countries, two criteria are established: (1) Relative to China, they are upstream in the GVC, primarily supplying technology and key components, but are not energy-exporting nations, and (2) They have close trade ties with China and serve as major export markets for Chinese goods. Based on data availability, the market-oriented countries include 14 nations: the United States, Japan, South Korea, the United Kingdom, Germany, France, the Netherlands, Switzerland, Luxembourg, Belgium, Austria, Italy, Portugal, and Spain. These countries complement China in the value chain, mainly responsible for R&D, technology development, and the production of core components. Furthermore, they are key export markets for Chinese finished goods. Between 2010 and 2016, exports to these 14 market-oriented countries accounted for nearly 50% of China's total exports, making them China's primary trading partners.

For competitive countries, two criteria are also set: (1) They have factor endowments similar to China, characterised by relatively low-cost labour, and (2) They maintain close economic and trade relations with market-oriented countries and have geographical advantages. Based on data availability, the selected competitive countries include 26 nations: Vietnam, Indonesia, Malaysia, the Philippines, and Thailand in Southeast Asia; India in South Asia; Mexico in Latin America; and Poland, Croatia, Hungary, Ukraine, Bulgaria, Latvia, the Czech Republic, Moldova, Bosnia and Herzegovina, Slovakia, Slovenia, Estonia, Belarus, Lithuania, Romania, and Serbia in Central and Eastern Europe; as well as Egypt, Morocco, and Tunisia in North Africa. During the period from 2010 to 2016, China's exports to these 26 countries represented 15% of its total export volume.

3.2 | Indicator Construction

3.2.1 | Construction of Core Explanatory Variables

Following the methodologies of Acemoglu and Restrepo (2020), Faber (2020), and Gihleb et al. (2022), this article constructs the penetration rate of robot usage to measure the extent of robot adoption. Based on the country classification in the core indicator design, three specific penetration rates are constructed: Penetration Rate of Robot Usage in Market-Oriented Countries Relative to China (*ETFRM*), Penetration Rate of Robot Usage in Competitive Countries Relative to China (*ETFCR*), and Domestic Penetration Rate of Robot Usage in China (*ETDR*). The construction methods are as follows:

$$ETFRM_{cit} = \ell_{cix}^M \frac{R_{it}^M O_{ix}^M}{L_{ix}^M}, ETFCR_{cit} = \ell_{cix}^C \frac{R_{it}^C O_{ix}^C}{L_{ix}^C}, ETDR_{cit} = \ell_{cit} \frac{R_{it}^{CN}}{L_{ix}^{CN}}$$

In the construction of indicators and subsequent empirical model design, this article redefines the subscripts and superscripts. The subscripts c , i , t and x represent city, industry, year and base year, respectively, with the base year set as $x=2010$. Setting the initial year of the sample period as the base year helps to avoid mutual influence between the robot variables and the weighting variables, thereby making the Bartik instrumental variable more reliable (Goldsmith-Pinkham et al. 2020). The superscript $j=\{M,C\}$ denotes the type of country, where M represents developed market-oriented countries, and C refers to other competitive countries.

Here, R_{it}^j represents the robot stock in industry i of country type j in year t , with data sourced from the IFR. O_{ix}^j denotes the share of imports from China in 2010 for industry i in country type j as a proportion of total demand, calculated as $O_{ix}^j = I_{ix}^{CNj} / E_{ix}^j$ using the 2010 World Input–Output Table. O_{ix}^j can be interpreted as the intensity of the relationship; without cross-border trade, the transmission effect of foreign robot usage on China would not occur. L_{ix}^j represents the number of workers in China's industry i in 2010 who are involved in producing export goods for country type j . Due to the lack of data on L_{ix}^j , this article defines $L_{ix}^j = f_{ix}^j L_{ix}$, where L_{ix} is the number of workers in China's industry i in 2010, calculated using the 2010 China Industrial Statistical Yearbook.⁴ f_{ix}^j is the ratio of China's exports in industry i to country type j relative to the total

exports of industry i , calculated using 2010 Chinese customs data.⁵ In summary, the economic interpretation of $\frac{R_{it}^j O_{ix}^j}{L_{ix}^j}$ is the density of 'affected' workers in China within industry i due to robot usage in country type j .

Additionally, $\ell_{cix}^j = L_{cix}^j / L_{cx}$ is the weight used to distribute robot density from the 'industry' level to the 'city-industry' level. Here, L_{cix}^j represents the number of workers in city c , industry i in 2010, producing export goods for country type j , and L_{cx} is the total number of workers in city c in 2010. Similarly, due to the lack of statistical data for L_{cix}^j , this article defines $L_{cix}^j = f_{cix}^j L_{cix}$, where L_{cix} is the number of workers in city c , industry i in 2010. Both L_{cix} and L_{cx} are calculated using the 2010 Population Census data.⁶ f_{cix}^j is the ratio of exports from city c to country type j relative to the total exports of city c , calculated using 2010 Chinese customs data.

It is worth noting that the indicator measuring the penetration rate of overseas robot adoption into the Chinese economy is based on estimation results. This is mainly due to the lack of data on the employment share of export-oriented production across cities and industries in China. As a result, this article can only use the corresponding export share as a proxy variable. Generally, export firms tend to have higher productivity, and the employment share engaged in export production is usually lower than the corresponding export share. Therefore, the core explanatory variable measured in this article may overstate the negative impact of overseas robot adoption on China's exports.

Finally, following the methodology of Gihleb et al. (2022), this article constructs the penetration rate of China's robot usage at the city-industry level, denoted as *ETDR*. To ensure that the test results are not driven by city- or industry-specific factors unrelated to robot usage, the model also includes several interaction terms for the penetration rate of foreign robot usage in China, referred to as the *main effects*. (1) The share of employment in city c , industry i , serving foreign demand: $major1_{cix}^j = \ell_{cix}^j = f_{cix}^j L_{cix} / L_{cx}$. (2) The degree of dependence of foreign countries on city c , industry i in China: $major2_{cix}^j = O_{ix}^j f_{cix}^j L_{cix} / L_{cx}$.

3.2.2 | Construction of Instrumental Variables

The construction process of the core indicators presents a reverse causality issue. Developed market-oriented countries, concerned about the widening trade deficit with China, may have started adopting robots to promote the 'reindustrialization' process. Additionally, China's decision to adopt robots could be influenced by firms' export intentions. The reverse causality problem may affect the reliability of the study's findings.

Drawing on the approach of Acemoglu and Restrepo (2020), this article selects a set of countries, referred to as 'instrumental variable countries', from outside the market-oriented and competitive countries. The robot stock of these 'instrumental variable countries' is used to replace domestic and foreign robot stocks, constructing instrumental variables to eliminate endogeneity. The selection of 'instrumental variable countries' must meet

two criteria: (1) they occupy a certain market position in the global robotics market, and (2) they have minimal economic and trade relations with China. Based on data availability, the selected ‘instrumental variable countries (IV)’ include 16 nations: Australia, New Zealand, Denmark, Sweden, Norway, Finland, Ireland, Iceland, Greece, Malta, Turkey, Israel, Russia, Canada, South Africa, and Brazil. From 2010 to 2016, China’s exports to these countries accounted for only 10% of China’s total exports, with many of these countries primarily focused on natural resource industries rather than traditional outsourcing industries. This article constructs instrumental variables for the penetration rate of robot usage in developed countries, competitive countries, and China as follows:

$$ETFRMiv_{cit} = \ell_{cix}^M \frac{R_{it}^{IV} O_{ix}^M}{L_{ix}^M}, ETFRCiv_{cit} = \ell_{cix}^C \frac{R_{it}^{IV} O_{ix}^C}{L_{ix}^C},$$

$$ETDRiv_{cit} = \ell_{cit} \frac{R_{it}^{IV}}{L_{ix}}$$

3.3 | Empirical Model Design

This article constructs the empirical model as shown in Equation (8):

$$\ln EXP_{cit} = \alpha + \beta_{fm} ETFRM_{cit} + \beta_{fc} ETFRC_{cit} + \beta_d ETDR_{cit} + Z_{cit}'\gamma' + \lambda_c + \lambda_i + \lambda_t + \varepsilon_{cit} \quad (8)$$

Here, c , i and t represent city, IFR industry and year, respectively.

3.3.1 | The Dependent Variable

$\ln EXP_{cit}$ is the logarithm of the export volume from city c , industry i , in year t . This article primarily examines the impact of industrial robot adoption in developed market-oriented countries on China’s export volume. For this purpose, customs data from 2010 to 2016 is aggregated to obtain export values at the city-industry level.⁷ On one hand, the total export volume at the ‘city-industry’ level in China is calculated; on the other hand, export values specifically to developed countries are also calculated based on the destination countries. Additionally, in the subsequent extension analysis, the export volume to countries other than the market-oriented countries is calculated to assess China’s capacity to explore new export markets.

3.3.2 | The Core Explanatory Variable

$ETFRM_{cit}$ represents the penetration rate of robot usage in developed countries. This article focuses on the impact of domestic supply in developed countries on China’s export volume following the adoption of robots by multinational enterprises’ home countries. Therefore, the penetration rate of robot usage in competitive countries, $ETFRC_{cit}$, and the domestic penetration rate of robot usage, $ETDR_{cit}$, are set as key control variables. Specifically, the foreign robot penetration rate represents the density of foreign robot usage mapped through the industrial chain to city c , industry i in China, while the domestic penetration rate indicates the density of robot usage within city c ,

industry i domestically. This article focuses on the size and direction of coefficient β_{fm} , as it critically reveals the extent to which the application of robotic technology affects China’s foreign trade.

3.3.3 | Time-Varying Variables at the City Level

Important time-varying variables at the city level, denoted as Z_{cit} , include the city’s gross domestic product (GDP), total population (pop), number of enterprises above a designated size ($firm$), highway freight volume ($sup1$), and number of international internet users ($sup2$). These time-varying variables are sourced from the China City Statistical Yearbook, and monetary variables are deflated using the GDP deflator. These time-varying variables significantly affect both the dependent variable and the core explanatory variables. Controlling for them can mitigate endogeneity to some extent.

3.3.4 | Fixed Effects and Robust Standard Error

City and industry fixed effects, λ_c and λ_i , are included to control for time-invariant factors at the provincial and industry levels. Additionally, year fixed effects, λ_t , account for disturbances that vary only over time. Finally, ε_{cit} represents robust standard errors clustered at the city level.

3.3.5 | Descriptive Statistics of Variables

This article matches each dataset using city administrative codes and industry codes, resulting in a city-industry panel dataset covering 292 cities and 11 industries from 2010 to 2016, totalling 21,770 observations. Table 2 provides the descriptive statistics of the main variables used in this article.

4 | Empirical Results

4.1 | Baseline Results

This article employs the two-stage least squares (2SLS) method with instrumental variables to estimate the baseline model in Equation (8). The core explanatory variable, ‘Penetration Rate of Industrial Robot Usage in Developed Countries Relative to China’, is treated as an endogenous variable, and corresponding instrumental variables are introduced to address endogeneity. Key control variables, including the ‘Domestic Industrial Robot Penetration Rate’ and the ‘Penetration Rate of Industrial Robot Usage in Competitive Countries Relative to China’, are also included in the estimation.

This article uses total export volume and exports to developed countries to represent the export performance at the ‘city-industry’ level in China. Columns (1) and (2) of Table 3 control only for the main effects and fixed effects, without accounting for time-varying city variables. The results show that the penetration rate of industrial robot usage in developed countries has a significant negative impact on various export performance measures at the ‘city-industry’ level, with coefficients of -3.58 and

TABLE 2 | Descriptive statistics of main variables.

Variables	Obs	Mean	SD	Min	Max
Total Exports	21,770	15.574	5.164	0	25.478
Exports to Market-Oriented Countries	21,770	13.533	6.246	0	24.429
Exports to New Markets	21,770	14.859	5.38	0	25.299
Domestic Robot Penetration Rate	21,770	0.103	0.351	0	12.688
Robot Penetration Rate in Market-Oriented Countries	21,770	0.048	0.22	0	7.162
Robot Penetration Rate in Competitive Countries	21,770	0.002	0.009	0	0.45
City GDP	20,114	20.591	0.868	18.372	23.51
Total Population of the City	20,130	10.548	0.645	7.55	12.734
Number of Enterprises Above a Designated Size in City	20,114	11.298	1.032	7.741	14.327
City Highway Freight Volume	20,026	13.585	0.852	10.576	17.191
Number of International Internet Users in City	19,930	17.744	0.898	15.413	22.367

−2.74, consistent with the findings for Mexico in Faber (2020). Columns (3) and (4) further control for time-varying city variables, resulting in an increased negative impact coefficient for the penetration rate of industrial robot usage in developed countries, which remains significant. These results provide preliminary evidence that the adoption of industrial robots in developed countries leads to lower production costs, with part of the demand being met through localised production, thereby reducing imports of Chinese goods.

The empirical research in this article partially confirms the conclusions derived from the theoretical model. First, the motivation for increased robot usage in market-oriented countries like Europe and the United States may stem from technological breakthroughs in robotics, in which case $\partial c_n^i / \partial O_n^{i*} < 0$. This results in a decline in production costs in developed countries,

reducing the competitive advantage of Chinese products in the international market and ultimately negatively impacting China's exports. Second, developed market-oriented countries, such as those in Europe and the United States, use robots to cope with rising labour costs in China. In this scenario, the elasticity of changes in production costs in developed countries is influenced by China, represented by $\Delta^j \rightarrow 0$. These two factors, combined with the theoretical discussions earlier, contribute to the decline in China's export volume.

Additionally, the domestic robot penetration rate has a significant positive effect on the export performance at the 'city-industry' level, which aligns with expectations (Graetz and Michaels 2018). This article explains it as follows: China's increased use of robots is primarily aimed at alleviating rising labour costs. Although the adoption of robots requires paying installation and operational costs, these costs are lower than those associated with hiring workers. Furthermore, the rise in China's production costs may lead to an increase in costs in other countries through cross-national and cross-industry linkages, potentially lowering China's relative costs and thereby boosting its export volume.

The penetration rate of industrial robot usage in competitive countries does not have a significant impact on export performance. This seems to confirm, to some extent, that the economic and trade relations between China and competitive countries are highly complex: some regions or sectors within China have a direct competitive relationship with competitive countries, while others maintain industrial chain cooperation. This complexity results in weaker statistical significance in the estimation results.

Comparing the estimation results of the variables *ETFRM* and *ETFRC*, we find that while the use of robots in developed countries has an adverse impact on China's exports, the impact from similar emerging countries using robots is not significant. This suggests that, in many sectors, China actually competes with more advanced developed countries and complements developing emerging countries. The findings indicate that, although more Western countries view China as a developed nation, China's industrial structure has not yet reached the level of developed countries and remains susceptible to shocks from them.

4.2 | Robustness Tests

4.2.1 | A Stricter Two-Stage Least Squares (2SLS) Method Is Applied

The key control variables, 'Domestic Industrial Robot Penetration Rate' and 'Penetration Rate of Industrial Robot Usage in Competitive Countries' also present endogeneity issues, so a more rigorous approach to handling endogeneity is necessary in this section. In Columns (1) and (2) of Table 4, the endogenous variables are set as 'Penetration Rate of Industrial Robot Usage in Developed Countries' and 'Domestic Industrial Robot Penetration Rate' with instrumental variables introduced for both. In Columns (3) and (4) of Table 4, the 'Penetration Rate of Industrial Robot Usage in Competitive Countries' is also treated as an endogenous variable, with corresponding instrumental variables

TABLE 3 | Baseline results.

	(1)	(2)	(3)	(4)
Variables	Total exports	Exports to market-oriented countries	Total exports	Exports to market-oriented countries
<i>ETFRM</i>	−3.575*** (1.283)	−2.735* (1.488)	−3.882*** (1.352)	−3.327** (1.610)
<i>ETDR</i>	1.146*** (0.377)	1.448*** (0.480)	1.327*** (0.381)	1.604*** (0.494)
<i>ETFRC</i>	−2.254 (2.203)	−4.503 (2.893)	−3.937** (1.933)	−6.070** (2.698)
City controls	No	No	Yes	Yes
Main effects	Yes	Yes	Yes	Yes
<i>KP rk LM</i>	17.124 (0.000)	17.124 (0.000)	17.584 (0.000)	17.584 (0.000)
<i>KP rk WF</i>	76.427 (16.380)	76.427 (16.380)	64.761 (16.380)	64.761 (16.380)
City FE	Yes	Yes	Yes	Yes
Industry × year FE	Yes	Yes	Yes	Yes
Observations number	21,616	21,616	19,842	19,842
<i>R</i> ²	0.008	0.010	0.018	0.017

Note: The values in parentheses below the coefficients are robust standard errors clustered at the city level. The values in parentheses below the *KP rk LM* statistics are the corresponding *p*-values, and the values in parentheses under the *KP rk WF* statistic represent the critical values for the Stock-Yogo weak instrument test at the 10% level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Due to space limitations, the estimation results for city time-varying variables and main effects are omitted. Unless otherwise stated, the following tables adhere to the same format.

introduced. After controlling for city variables, main effects, and fixed effects, the study finds that the impact of ‘Penetration Rate of Industrial Robot Usage in Developed Countries’ on export volumes at the ‘city-industry’ level remains stable, consistent with the baseline results, confirming the robustness of the study’s conclusions. Although the introduction of more instruments in columns (3) and (4) leads to potential issues of weak instrument identification, the inclusion of additional instruments allows for a more accurate assessment of the impact of robot adoption in the domestic economy and in competing countries, thereby partially validating the baseline results.

4.2.2 | Excluding Exports to Robot-Exporting Countries

The world’s main robot manufacturers are concentrated in Germany, Japan and South Korea, which supply the majority of robots globally (IFR 2019). These robot-exporting countries hold a greater degree of control in localised production, and their decisions to supply robots may influence bilateral trade, thereby increasing the endogeneity between robot usage in developed countries and China’s exports. On one hand, if robot-exporting countries deem it profitable to produce in the current host country, they may supply more robots to enhance productivity, mitigating the host country’s exposure to the impact of reindustrialization in market-oriented countries. On the other hand, if a robot-exporting country intends to adjust trade relations, whether by switching to a new host country or achieving

self-sufficiency, it can reshape its production capacity by changing the direction of robot supply.

In China, both export volume and robot usage are significantly influenced by robot-exporting countries, with the majority of robot supplies coming from Japanese manufacturers. In reports from the International Federation of Robotics prior to 2013, data on robots in China was primarily provided by the Japan Robot Association and Japanese-invested robot suppliers in China. Based on this, the study excludes China’s exports to the world’s major robot-exporting countries from the total export volume and the export volume to developed countries to avoid interference in the analysis. The empirical results are shown in Columns (1) and (2) of Table 5. After excluding exports to Germany, Japan, and South Korea, the study finds that the impact of the penetration rate of robot usage in developed countries on export performance at the ‘city-industry’ level remains stable.

4.2.3 | Controlling for China’s Labor Factor Costs

The primary characteristic that distinguishes industrial robots from ordinary capital goods is their complete substitution for human labour. While ordinary capital goods can replace some low-skilled labour to achieve higher efficiency and quality, they still require workers to operate the machinery. This operation process has a higher technical requirement, necessitating significant investment in technical training and higher wages for

TABLE 4 | Stricter two-stage least squares (2SLS) method.

	(1)	(2)	(3)	(4)
Variables	Total exports	Exports to market-oriented countries	Total exports	Exports to market-oriented countries
<i>ETFRM</i>	−2.652*** (0.739)	−2.738*** (0.898)	−2.845*** (0.968)	−2.527** (1.211)
<i>ETDR</i>	0.557* (0.327)	1.236*** (0.437)	0.713* (0.389)	1.066** (0.517)
<i>ETFRC</i>	0.414 (2.241)	−3.988 (2.870)	−0.990 (3.478)	−2.457 (4.864)
City controls	Yes	Yes	Yes	Yes
Main effects	Yes	Yes	Yes	Yes
<i>KP rk LM</i>	12.990 (0.000)	12.990 (0.000)	7.275 (0.007)	7.275 (0.007)
<i>KP rk WF</i>	30.139 (7.030)	30.139 (7.030)	6.773 (NA)	6.773 (NA)
City FE	Yes	Yes	Yes	Yes
Industry × year FE	Yes	Yes	Yes	Yes
Observations number	19,842	19,842	19,842	19,842
<i>R</i> ²	0.023	0.018	0.023	0.019

Note: In Columns (1) and (2), 'Penetration Rate of Industrial Robot Usage in Market-Oriented Countries' and 'Domestic Industrial Robot Penetration Rate' are treated as endogenous variables. In Columns (3) and (4), 'Penetration Rate of Industrial Robot Usage in Market-Oriented Countries', 'Domestic Industrial Robot Penetration Rate', and 'Penetration Rate of Industrial Robot Usage in Competitive Countries' are all treated as endogenous variables. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

workers (Humlum 2019). In contrast, robots can perform automated operations in the production process through programming, enabling a more thorough replacement of human labour (Frey and Osborne 2017).

Given the attribute of robots as substitutes for labour, the labour factor costs in the host country become a key determinant of robot usage in developed countries. When labour costs in a region increase, it directly affects the local labour market structure, robot adoption rate, and export trade, potentially interfering with the study's results. To address this, the study includes city-level average wage levels and monthly minimum wage levels in the city time-varying variables. The city average wage data are sourced from the China City Statistical Yearbook, and the minimum wage data come from the China Research Data Service Platform (CNRDS).⁸ Both are deflated using the Consumer Price Index (CPI) and then logarithmically transformed. The results are shown in Columns (3) and (4) of Table 5. Even after controlling for city labour factor costs, the study still finds a stable impact of foreign robot penetration rates on export performance at the 'city-industry' level.

4.2.4 | Controlling for City-Year Interaction Fixed Effects

When using the Bartik instrumental variable, a key assumption is that the fixed employment share ℓ_{civ} , which drives differences

in domestic and foreign robot penetration rates at the city level, is uncorrelated with the random error term (Goldsmith-Pinkham et al. 2020). However, in constructing the instrumental variables, it has not been rigorously demonstrated that the fixed employment share is independent of the random error term. To address this potential threat, this article follows the approach of Faber (2020) by controlling for 'city-year' interaction fixed effects and 'industry-year' interaction fixed effects in the baseline model. This allows for a strict control of city-specific time-varying disturbances and industry-specific time-varying disturbances.⁹ After including these two interaction fixed effects, all city time-varying variables are removed, and the empirical results are shown in the first two columns of Table 6. Additionally, the study further controls for 'province-industry-year' triple interaction fixed effects to account for more regional and industry-level disturbances, with the results shown in the last two columns of Table 6. The findings indicate that the penetration rate of industrial robot usage in developed countries continues to have a highly significant negative impact on exports at the 'city-industry' level, suggesting that the impact of robot adoption on China's exports is robust.

4.2.5 | Validity Test of the Bartik Method

This article employs the Bartik method to construct the indicator of overseas robot adoption penetration into the Chinese economy and uses it as an explanatory variable. To obtain consistent estimation results, the initial share weights at the city level must

TABLE 5 | Controlling for robot-exporting countries and China's labour costs.

	(1)	(2)	(3)	(4)
Variables	Total exports excluding Japan, South Korea and Germany	Exports to market- oriented countries excluding Japan, South Korea and Germany	Total exports	Exports to market- oriented countries
<i>ETFRM</i>	−3.992*** (1.378)	−4.508** (1.779)	−3.878*** (1.352)	−3.282** (1.606)
<i>ETDR</i>	1.363*** (0.390)	1.899*** (0.547)	1.317*** (0.381)	1.573*** (0.493)
<i>ETFRC</i>	−4.003** (1.986)	−7.181** (3.047)	−3.865** (1.936)	−5.938** (2.696)
City average wage			−0.941 (0.646)	−1.375* (0.829)
City minimum wage			0.219 (0.472)	0.151 (0.558)
City controls	Yes	Yes	Yes	Yes
Main effects	Yes	Yes	Yes	Yes
<i>KP rk LM</i>	17.584 (0.000)	17.584 (0.000)	17.455 (0.000)	17.455 (0.000)
<i>KP rk WF</i>	64.761 (16.380)	64.761 (16.380)	63.925 (16.380)	63.925 (16.380)
City FE	Yes	Yes	Yes	Yes
Industry × year FE	Yes	Yes	Yes	Yes
Observations number	19,842	19,842	19,529	19,529
R^2	0.018	0.013	0.019	0.018

Note: *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

satisfy the exogeneity condition, meaning they should not be correlated with the error term that may affect China's exports. However, this exogeneity assumption is difficult to test directly. Therefore, this article evaluates its validity through a series of alternative estimation methods.

First, this article analyses the correlation of share weights. Specifically, it examines the correlation between the initial industry shares of the four cities with the largest weights and their initial city characteristics. The industries with higher share weights include apparel and textiles, chemicals and chemical engineering, metal machinery, and electronics and electrical equipment, while the city characteristics considered include gross domestic product (*GDP*), total population (*pop*), the number of large-scale enterprises (*firm*), highway freight volume (*sup1*), and the number of international internet users (*sup2*). The correlation analysis results are presented in Columns (1) to (4) of Table 7. Overall, the R^2 values of these correlation tests are relatively low, and most estimated coefficients do not reach statistical significance. This suggests that, under certain conditions, the share weights can be considered exogenous.

Second, this article examines the pre-trend of the share weights. Following the approach of Acemoglu and Restrepo (2020), the study matches China's export value from 2000 to 2006 with the robot adoption exposure index from 2010 to 2016 and repeats the tests conducted in the baseline regression. The results are presented in Columns (5) and (6) of Table 7. The regression results show that the estimated coefficients of foreign robot adoption exposure are not statistically significant. This indicates that the share weights do not exhibit a systematic trend in the pre-period, thereby supporting the validity of the Bartik method.

5 | Extension Analysis

Developed market-oriented countries use industrial robots as a means to counter rising labour costs in China, which reduces China's export performance. However, there are significant differences across industries in terms of raw material input, trade facilitation under international division of labour, and levels of robot technology application. These industry-level differences lead to varying impacts of robot usage in

TABLE 6 | Changing fixed effects.

	(1)	(2)	(3)	(4)
Variables	Total exports	Exports to market-oriented countries	Total exports	Exports to market-oriented countries
<i>ETFRM</i>	−3.964*** (1.311)	−2.992** (1.516)	−3.123*** (1.173)	−3.400*** (1.274)
<i>ETDR</i>	1.464*** (0.401)	1.698*** (0.521)	1.188*** (0.396)	1.459*** (0.438)
<i>ETFRC</i>	−3.877* (2.246)	−5.751* (3.145)	−3.977 (2.736)	−5.828* (3.253)
City controls	No	No	Yes	Yes
Main effects	Yes	Yes	Yes	Yes
<i>KP rk LM</i>	17.435 (0.000)	17.435 (0.000)	20.021 (0.000)	20.021 (0.000)
<i>KP rk WF</i>	79.403 (16.380)	79.403 (16.380)	89.695 (16.380)	89.695 (16.380)
City × year FE	Yes	Yes	No	No
Industry × year FE	Yes	Yes	No	No
Prov × industry × year FE	No	No	Yes	Yes
Observations number	21,616	21,616	19,488	19,488
<i>R</i> ²	0.008	0.012	0.210	0.206

Note: *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

developed countries on China's exports. This section further examines the moderating roles of cross-industry intermediate input, international trade barriers, and industry robotisation levels.

5.1 | Cross-Industry Input Share and International Trade Costs

The production process of enterprises relies on the input of intermediate goods, which may come from other industries or other countries. In the aforementioned Proposition 2: First, we believe that the use of robots will cause changes in the production costs of the domestic industry, with the direction determined by the sign of $\partial c_n^j / \partial O_n^{j*}$; second, along with the inter-industry and cross-country inputs in the production network, the cost changes of the domestic industry will be transmitted to other countries and other industries in the same direction, determined by $\partial c_i^k / \partial c_n^j$ (equivalent to the impact of the use of robots in the domestic industry on the production costs of other industries in other countries, denoted as $\partial c_i^k / \partial O_n^{j*}$); finally, the cost changes in other countries and other industries caused by the use of robots will be transmitted back to the domestic industry through the inter-industry and cross-country network, affecting $\partial c_n^j / \partial O_n^{j*}$ again. As shown in Figure 1.

In the process where the production costs of industry j in country n interact with those of industry k in country i , the dependence

of industry j in country n on the input share γ_n^{kj} from industry k , as well as the trade cost κ_{in}^j that country i provides to country n for industry j 's intermediate goods, are key in determining the impact. Proposition 2 states that the larger γ_n^{kj} is and the smaller κ_{in}^j is, the more significant the production cost changes caused by the use of robots will be through the associated effects. Conversely, the impact will be less significant. The following text attempts to test the moderating effects of inter-industry input shares and international trade costs.

5.1.1 | The Moderating Role of Inter-Industry Input Shares

This article uses the World Input–Output Database (WIOD) to calculate the inter-industry input shares for various industries. First, we select the world input–output tables for the years 2010 and 2014. The sample corresponds to the 2010 input–output table for the years 2010–2013, and the 2014 input–output table for the years 2014–2016, calculating the inter-industry input shares for the respective years. Second, we match the WIOD industries with the IFR industries, calculate the total intermediate input for each country and industry, and determine the proportion of inter-industry and cross-country inputs, as well as the proportion of cross-country inputs in total inputs, denoted as s_{njt}^T . Here, $T \in \{k, h, g, kg\}$ represents both inter-industry and cross-country, as well as solely cross-country inputs. Third, we select 13 countries that are classified as developed market economies, sum the total intermediate

TABLE 7 | Validity test of the Bartik method.

	(1)	(2)	(3)	(4)	(5)	(6)
Variables	Apparel and textiles	Chemicals and chemical engineering	Metal and machinery	Electronics and electrical equipment	Total exports (2000–2006)	Exports to market-oriented countries (2000–2006)
<i>GDP</i>	−0.091* (0.008)	0.009 (0.009)	0.018 (0.041)	0.015 (0.028)		
<i>pop</i>	0.010 (0.014)	−0.002 (0.002)	0.014* (0.002)	−0.009 (0.007)		
<i>firm</i>	0.014** (0.001)	0.002** (0.000)	0.006 (0.001)	0.004** (0.000)		
<i>sup1</i>	−0.042 (0.018)	−0.010 (0.005)	−0.043 (0.012)	−0.020 (0.019)		
<i>sup2</i>	0.019 (0.011)	−0.002 (0.002)	−0.002 (0.008)	−0.006 (0.009)		
<i>ETFRM</i>					−1.502 (1.254)	−0.589 (1.305)
<i>ETDR</i>					0.816* (0.409)	0.609 (0.448)
<i>ETFRC</i>					0.473 (2.079)	1.426 (2.276)
City controls					Yes	Yes
Main effects					Yes	Yes
<i>KP rk LM</i>					17.584 (0.000)	17.584 (0.000)
<i>KP rk WF</i>					64.761 (16.380)	64.761 (16.380)
City FE					Yes	Yes
Industry × year					Yes	Yes
Observations	205	207	204	207	19,842	19,842
<i>R</i> ²	0.357	0.366	0.230	0.309	0.011	0.015

Note: *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

inputs for industry j across these 13 countries, and calculate the intermediate input share for industry j in country n . Using this share as the weight, we aggregate to obtain the cross-industry and cross-country input share, as well as the cross-country input share for industry j in developed countries, as shown below: $S_{jt}^T = \sum_{n \in M} \left(\text{input}_{njt} / \sum_{n \in M} \text{input}_{njt} \right) s_{njt}^T$, where M represents the set of developed market countries.¹⁰ Finally, based on the cross-industry and cross-country input shares in developed countries, we use the median as the critical value. Industries with a share greater than or equal to the critical value are labelled as ‘high cross-border input industries’, and the corresponding dummy variable is set to 1; otherwise, the

dummy variable is set to 0, transforming $S_{jt}^T \in \{k\text{hg}, \text{kg}\}$ into the corresponding binary dummy variable.

This article takes the robot usage penetration rate *ETFRM* of developed market economies as the core explanatory variable of endogeneity and introduces corresponding instrumental variables. In addition, this article also includes the interaction term between the robot usage penetration rate of developed market economies and the binary dummy variable to test the moderating effect. Correspondingly, this article uses the interaction of the instrumental variable of the robot usage penetration rate in developed market economies and the binary dummy variable as the instrumental variable for the interaction term. The model is shown in

the following formula, where c, j , and t represent cities, industries, and years, respectively. The coefficient β_{jh} of the interaction term indicates the moderating effect of the binary dummy variable S_{jt}^T on the impact of robot usage in market-type countries.

$$\ln EXP_{cit} = \alpha + \beta_{jh} ETFRM_{cit} \times S_{jt}^T + \beta_{fm} ETFRM_{cit} + \beta_{fc} ETFRC_{cit} + \beta_d ETDR_{cit} + Z_{ct} \gamma' + \lambda_c + \lambda_i + \lambda_t + \varepsilon_{cit}$$

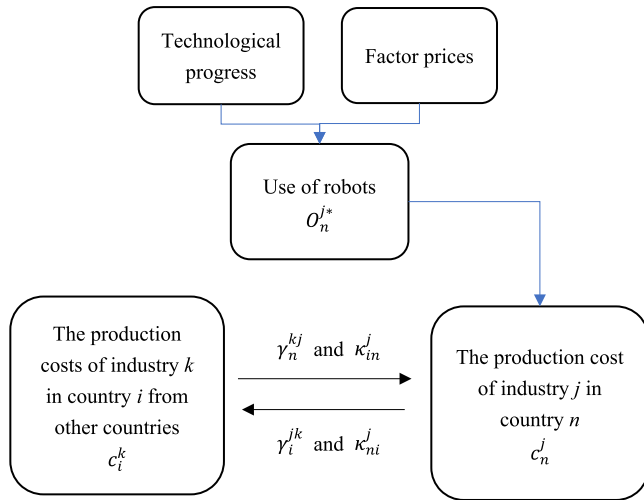


FIGURE 1 | The impact mechanism of cross-industry input share. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

The empirical results are shown in Table 8. In columns (1) and (2) of Table 8, the binary dummy variable S_{jt}^T is ‘whether it is a high cross-border input industry (khg)’, and the estimation is conducted using the two-stage least squares method. It can be found that the coefficient β_{jh} of the interaction term is significantly negative, indicating that under a higher cross-industry and cross-country input share, the negative impact of robot usage in market-type countries on China’s ‘city-industry’ export scale will be intensified. This conclusion is consistent with the speculation of Proposition 2 in this article, that is, the more dependent the production of products within the industry is on cross-industry intermediate inputs, the greater the impact of robot usage in that industry will be through the industry association effect. Since this article studies the impact of robot usage in foreign developed countries on China’s exports, therefore, this impact will only occur in cross-country and cross-industry inputs. Furthermore, we add the core explanatory variable and the interaction term with ‘whether it is a high cross-country input industry (kg)’, and the estimation results are shown in columns (3) and (4). It can be found that

TABLE 8 | The moderating effect of inter-industry input shares.

Variables	(1)	(2)	(3)	(4)
	Total exports	Exports to market-oriented countries	Total exports	Exports to market-oriented countries
$ETFRM \times khg$	−1.223*** (0.387)	−1.158** (0.499)		
$ETFRM \times kg$			−1.280*** (0.472)	−1.380** (0.549)
$ETFRM$	−0.304*** (0.102)	−0.253** (0.121)	0.885** (0.441)	1.039** (0.520)
$ETDR$	1.511*** (0.462)	1.777*** (0.586)	1.408*** (0.398)	1.691*** (0.512)
$ETFRC$	−3.841* (2.086)	−5.978** (2.936)	−4.418** (2.004)	−6.587** (2.724)
City controls	Yes	Yes	Yes	Yes
Main effects	Yes	Yes	Yes	Yes
$KP rk LM$	31.889 (0.000)	31.889 (0.000)	17.720 (0.000)	17.720 (0.000)
$KP rk WF$	213.902 (7.030)	213.902 (7.030)	32.855 (7.030)	32.855 (7.030)
City FE	Yes	Yes	Yes	Yes
Industry $\times$ year FE	Yes	Yes	Yes	Yes
Observations number	19,842	19,842	19,842	19,842
R^2	0.015	0.013	0.018	0.016

Note: *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

the negative impact of robot usage in developed countries is mainly caused by cross-country inputs.

5.1.2 | The Moderating Role of International Trade Costs

This article utilises the World Integrated Trade Solution (WITS) to calculate the export tariff situation for various industries in China. First, we obtain the product-level export tariff rate data for China from 2010 to 2016 for 162 countries or regions worldwide and match the 4-digit HS codes to IFR industry codes. Second, we aggregate the product tariffs for each year to the IFR industry level, weighted by trade volume, as shown in the following formula:

$$\tau_i^j = \sum_{n \in N} \sum_{\omega^j \in \Omega^j} \left(e_{in}^{\omega^j} / \sum_{n \in N} \sum_{\omega^j \in \Omega^j} e_{in}^{\omega^j} \right) \tau_{in}^{\omega^j}$$

ω^j represents products ω in industry j ; Ω^j represents the set of all products within industry j ; $\tau_{in}^{\omega^j}$ represents the weighted tariff for product ω^j exported from China i to country n , based on the effective applicable tariff published by WITS; $e_{in}^{\omega^j}$ represents the export volume of product ω^j from China i to country n ; N represents the set of countries importing Chinese products: on the one hand, we set all 162 countries or regions in the data as the country set N , and on the other hand, we only set 14 market-oriented countries as

the country set N , thus constructing tariff rates for Chinese export industries categorised as ‘all countries’ and ‘market-oriented countries’. Finally, we convert the above industry-level tariffs into a binary dummy variable *Tariff*, using the median as the critical value. Industries with a tariff greater than or equal to the critical value are labelled as ‘high trade cost industries’, with the corresponding dummy variable *Tariff* set to 1; otherwise, it is set to 0.

Consistent with the approach in the previous text, this article incorporates the interaction term between the robot usage penetration rate of developed market-type countries and a binary dummy variable into the baseline model, treating the interaction term and the robot usage penetration rate of developed market-type countries as endogenous variables, and estimates using the two-stage least squares method. We control for the three-way interaction fixed effects of ‘province-industry-year’ to address the impact of the binary dummy variable *Tariff* as an industry time-varying variable. The estimation results are presented in Table 9. Columns (1) and (2) of Table 9 show the estimation results when calculating *Tariff* with all countries as N , and columns (3) and (4) show the results when calculating *Tariff* with 14 market-type countries as N . For both methods, the interaction term coefficients are significantly positive, which validates the conjecture in Proposition 2 regarding high trade costs weakening the impact of robot usage in market-type countries. This is in line with the actual situation; when the trade cost κ_{in}^j for industry j between China i and market-type country n is excessively high,

TABLE 9 | The moderating role of international trade costs.

Variables	(1)	(2)	(3)	(4)
	All counties		Market-oriented countries	
	Total Exports	Exports to market-oriented countries	Total exports	Exports to market-oriented countries
<i>ETFRM</i> × <i>tariff</i>	1.011*** (0.382)	1.620*** (0.530)	0.889* (0.493)	0.941 (0.610)
<i>ETFRM</i>	−3.269*** (1.174)	−3.635*** (1.277)	−2.930** (1.178)	−3.197** (1.281)
<i>ETDR</i>	0.985** (0.408)	1.134** (0.440)	1.136*** (0.393)	1.404*** (0.433)
<i>ETFRC</i>	−3.570 (2.691)	−5.176 (3.190)	−3.796 (2.718)	−5.637* (3.231)
City controls	Yes	Yes	Yes	Yes
Main Effects	Yes	Yes	Yes	Yes
<i>KP rk LM</i>	21.892 (0.000)	21.892 (0.000)	21.766 (0.00)	21.766 (0.00)
<i>KP rk WF</i>	55.717 (7.030)	55.717 (7.030)	47.840 (7.030)	47.840 (7.030)
Prov × industry × year FE	Yes	Yes	Yes	Yes
Observations number	19,488	19,488	19,488	19,488
<i>R</i> ²	0.211	0.207	0.211	0.207

Note: *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

it indicates the presence of protective policies in the market-type country in that field, and its initial bilateral trade scale π_{in}^j may be relatively small. Consequently, even if market-type countries start using industrial robots, the impact on China's trade is relatively limited.

5.2 | Industry Robotisation Level

Regarding the degree of robotisation across industries, there is currently no well-established measurement method. Therefore, this article indirectly examines the impact of industry robotisation through three approaches:

5.2.1 | Global Value Chain Participation

Generally, in country n , the more upstream industry j is in the GVC, the lower the degree of robotisation for production tasks within country n , while the higher the degree of robotisation for outsourced production tasks outside country n .¹¹ Countries positioned upstream in the GVC typically engage in activities such as technological R&D, product marketing, and the supply of core components, which require more high-skilled workers and are less substitutable by robots.

Based on this, following Koopman et al. (2010), this article uses WIOD data to calculate four indicators at the national and industry levels from 2010 to 2014: total GVC participation, forward participation, backward participation, and GVC position.¹² Given the study's focus on the impact of industrial robot usage in foreign developed countries on China's export performance, only developed market-oriented countries are selected, and the national average of each GVC indicator is calculated annually at the industry level: $GVC_{jt}^T = \sum_{n=1}^M gvc_{njt}^T / M$. Here, $T = \{tp, bp, fp, pt\}$, represent total participation, backward participation, forward participation, and

GVC position, respectively, and M is the number of developed countries. Similarly, this article incorporates interaction terms into the baseline model to construct the empirical model shown below, testing the moderating effect of industry robotisation levels. The model includes a 'province-industry-year' triple interaction fixed effect, λ_{pjt} , to control for the independent GVC_{jt}^T term. The coefficient β_{jh} preceding the interaction term captures the moderating effect of industry robotisation levels.

$$\ln EXP_{cjt} = \alpha + \beta_{jh} ETRM_{cjt} \times GVC_{jt}^T + \beta_{fm} ETRM_{cjt} + \beta_{fc} ETRC_{cjt} + \beta_d ETRD_{cjt} + Z_{ct} \gamma' + \lambda_{pjt} + \varepsilon_{cjt}$$

First, as shown in Table 10, Panel A, the negative impact of increased robot usage in foreign developed countries on China's export trade is stronger in industries with higher value chain participation. This result supports Proposition 1: higher GVC participation implies a larger cross-industry input share γ_n^{kj} and lower international trade costs κ_{in}^j . Through the effects of input-output linkages and international trade, the cost reduction achieved by robots becomes more pronounced, thereby amplifying the negative impact on trade.

Second, as shown in Table 10, Panels B and C, in developed countries, industries with higher forward and backward participation rates experience a more pronounced negative impact on China's export trade when they increase robot usage. Additionally, the amplifying effect of forward participation is greater than that of backward participation. Forward participation represents the share of total exports attributable to indirect value-added, while backward participation represents the share of total exports attributable to foreign value-added. Therefore, higher forward participation indicates a position upstream in the GVC. When industry j in country n is positioned upstream in the value chain, it implies that highly robotisable downstream jobs are outsourced to developing country i , with the outsourcing share approximately equal to $O_n^{j*} - \varpi_n^j + 1$. The farther upstream the industry is, the larger

TABLE 10 | Industry global value chain participation and position.

Dependent variables	β_{fm}	SD	β_{jh}	SD	Obs	KP rk LM	KP rk WF
Panel A: Total participation							
Total Exports	−3.367***	0.691	−2.370	0.456	19,488	58.495	60.111
Exports to Market-Oriented Countries	−4.090***	0.820	−2.969	0.554	19,488	58.495	60.111
Panel B: Backward participation							
Total Exports	−2.901***	0.624	−3.768	0.776	19,488	64.233	170.246
Exports to Market-Oriented Countries	−3.470***	0.732	−4.657	0.929	19,488	64.233	170.246
Panel C: Forward participation							
Total Exports	−3.362***	0.722	−4.881	0.946	19,488	45.992	40.808
Exports to Market-Oriented Countries	−4.129***	0.860	−6.201	1.167	19,488	45.992	40.808
Panel D: GVC position							
Total Exports	−1.114***	0.406	−2.197	0.964	19,488	29.951	84.289
Exports to Market-Oriented Countries	−1.214***	0.454	−2.400	1.111	19,488	29.951	84.289

Note: The p -values corresponding to the KP rk LM statistic are all '0.000', and the critical values for the Stock-Yogo weak instrument test at the 10% level corresponding to the KP rk WF statistic are all '7.030'. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

this interval becomes. Thus, when robots are used, industries positioned upstream in the value chain or with higher forward participation experience more significant effects.

Finally, as presented in Table 10, Panel D, in industries with higher positions in the GVC, the increase in robot usage in foreign developed countries has a more pronounced negative impact on China's export trade. This finding further indirectly supports Proposition 2: if an industry in a foreign developed market-oriented country is positioned upstream in the value chain, it primarily outsources downstream processes in the value chain. When robots are used, the impact is mainly felt in these downstream processes. The further downstream the industry, the greater the robot-affected share $O_n^{j*} - \omega_n^j + 1$. Therefore, if foreign developed market-oriented countries are positioned upstream in the value chain relative to China, then as China's labour cost advantage gradually diminishes, the increase in robot usage by these developed countries to perform downstream production leads to a more significant $\partial c_n^j / \partial O_n^{j*} < 0$, amplifying the negative impact on China's exports.

5.2.2 | Impact on Intermediate Goods or Final Goods Trade

Next, this article classifies products into intermediates and final goods to analyse the heterogeneous impact of robot usage in developed market-oriented countries on China's exports of different product types. Upstream segments of the supply chain primarily produce intermediates, while downstream segments

produce final goods. If the impact of robot usage in developed countries is more pronounced in downstream production stages, then China's exports of final goods will experience a more significant decline. The analysis in this section is as follows: First, six-digit HS codes are obtained from the customs database, and all HS codes are standardised to the 2012 version. Second, a concordance table between the 2012 HS codes and BEC codes is obtained from the UN Trade Statistics database to classify all export transactions into intermediate and consumer goods exports.¹³ Finally, exports are aggregated by country and product type, with the corresponding export performance indicators serving as the dependent variables. After transforming the dependent variables, the baseline estimation is repeated, with results presented in Table 11.

It can be observed that the increase in industrial robot usage in developed market-oriented countries has a significant negative impact on the 'total export volume' and 'exports to market-oriented countries' of final goods, indicating that the shock mainly affects China's exports of final goods. However, while the estimated coefficients for various export performance indicators of intermediates are also negative, the magnitude of the impact is relatively small and not statistically significant, suggesting that China's intermediate exports are not noticeably affected by the increased use of industrial robots in developed countries. In previous analyses, the moderating effect of the position of industries within the GVC in developed countries indicated that the impact of robot usage is more concentrated in downstream stages of the supply chain. These downstream stages involve the input of intermediates to complete the production and processing

TABLE 11 | Heterogeneity by product category.

Variables	Total exports		Exports to market-oriented countries	
	Intermediates	Final goods	Intermediates	Final goods
	(1)	(2)	(3)	(4)
<i>ETFRM</i>	-1.294 (1.149)	-2.436* (1.289)	-0.378 (1.524)	-3.327** (1.466)
<i>ETDR</i>	0.669** (0.338)	1.498*** (0.476)	0.884* (0.469)	1.250*** (0.448)
<i>ETFRC</i>	-1.825 (1.934)	-7.332* (3.848)	-2.625 (2.992)	-6.872** (3.361)
City controls	Yes	Yes	Yes	Yes
Main effects	Yes	Yes	Yes	Yes
<i>KP rk LM</i>	17.584 (0.000)	17.584 (0.000)	17.584 (0.000)	17.584 (0.000)
<i>KP rk Wald F</i>	64.761 (16.380)	64.761 (16.380)	64.761 (16.380)	64.761 (16.380)
City FE	Yes	Yes	Yes	Yes
Industry $\times$ year FE	Yes	Yes	Yes	Yes
Observations number	19,842	19,842	19,842	19,842
R^2	0.018	0.021	0.016	0.014

Note: *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

of consumer goods, which often require lower levels of technical complexity and are thus more likely to be performed by robots in routine tasks. Consequently, the impact on consumer goods is more pronounced. Moreover, capital goods and intermediates usually embody core value-added and technological difficulty within the production chain, necessitating higher international collaboration and skilled labour. This finding further indirectly supports Proposition 2: the greater the degree of robotisability of an industry, the stronger the negative impact of robot usage on China's exports.

5.2.3 | Industry Wage Levels

The impact of robot adoption in developed countries on China's industry-level exports is constrained by the degree of industry robotisation. Inspired by the robot substitution effect, the labour structure characteristics of an industry are a key determinant of differences in robotisation potential. Robots primarily replace simple and routine tasks, making industries with a higher share of such jobs more susceptible to the impact of robot adoption in developed countries. Moreover, since workers in simple and routine jobs typically earn lower wages, a lower average industry wage level reflects a higher degree of robotisation potential.

To analyse this, this article utilises 2007 industrial enterprise data to calculate firm-level average wages based on the number of employees and total payable wages reported by firms. After removing extreme values, the industry wage level (*wage*) is obtained by taking the simple average of firm-level wages within each industry. Subsequently, the industry wage level is interacted with *ETFRM* and *ETFRiv* to construct interaction terms and their instrumental variables, which are then incorporated into the econometric model to replicate the baseline regression. The results, presented in Table 12, show that the interaction term coefficients are significantly positive, indicating that in industries with lower wage levels, the negative impact of robot adoption in developed countries on China's exports is more pronounced. This finding aligns with expectations: industries with higher average wages tend to have a lower share of simple and routine tasks, making them less robotisable and thus less affected by the robot substitution effect.

5.3 | Other Extension Analyses

The use of robots in developed market-oriented countries may create a 're-offshoring' shock for China, leading to a decrease in export volume. This section conducts further extension analyses in two areas: first, examining industry heterogeneity in the impact; second, exploring whether the use of robots in developed market-oriented countries affects China's efforts to develop new export markets.

5.3.1 | Industry Heterogeneity

The impact of robot usage in developed countries varies across industries. On the one hand, each industry's level of trade participation differs, with China's exports primarily concentrated in the 'apparel and textiles' and 'electronics, electrical

TABLE 12 | Heterogeneous impact of industry average wage levels.

Variables	(1)	(2)
	Total exports	Exports to market-oriented countries
<i>ETFRM</i>	−4.209** (1.857)	−4.030* (2.200)
<i>ETFRM</i> × <i>wage</i>	8.585** (3.789)	8.223* (4.491)
<i>ETDR</i>	1.459*** (0.421)	1.730*** (0.540)
<i>ETFRC</i>	−4.036** (2.006)	−6.165** (2.860)
City controls	Yes	Yes
Main effects	Yes	Yes
<i>KP rk LM</i>	22.100 (0.000)	22.100 (0.000)
<i>KP rk WF</i>	47.592 (7.030)	47.592 (7.030)
City FE	Yes	Yes
Industry × year FE	Yes	Yes
Observations number	19,842	19,842
<i>R</i> ²	0.012	0.014

Note: *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

appliances, and instrumentation manufacturing' sectors. The deeper the participation, the stronger the negative impact. On the other hand, the applicability of industrial robots varies across industries; in some industries, such as automotive manufacturing, the majority of positions can be replaced by industrial robots, whereas other industries still rely heavily on human labour. Based on this, the total sample is divided into 11 industry subsamples according to IFR industry classifications, and 2SLS estimation is conducted for each subsample, controlling for city time-varying variables, as well as city and year fixed effects, to examine the differential impact of robot usage in developed market-oriented countries on various industries in China.

The estimation results are presented in Table 13. In Table 13, Column (1) lists the industry codes for the current sample, and the coefficient β_{fm} represents the estimated coefficient of 'Penetration Rate of Robot Usage in Developed Countries'. Additionally, the table displays standard errors clustered at the city level, *t*-values, number of observations, and the *KP* statistic. The dependent variable is the total export volume at the city level in China. The results indicate substantial variation in the impact of robot usage penetration in developed countries on China's export performance, with a significantly negative effect observed only in the 'electronics, electrical appliances, and instrumentation manufacturing' industry.

TABLE 13 | Industry heterogeneity.

Industry Desc	β_{fm}	SD	Obs	KP rk LM	KP rk WF
Food and beverages	4.122	2.557	1819	40.409 (0.000)	7.4e+04
Apparel and textiles	−2.018	7.954	1819	28.839 (0.000)	678.442
Wood and furniture	0.403	0.413	1812	9.672 (0.002)	960.524
Paper and printing	1.060**	0.476	1784	6.971 (0.008)	4.288
Chemicals and chemical products	−4.280	2.996	1832	4.552 (0.033)	78.701
Glass and cement	1.811	2.411	1825	2.331 (0.127)	2.312
Metal and machinery	−0.462	0.661	1832	38.005 (0.000)	586.291
Automotive manufacturing	−0.528	0.510	1791	27.888 (0.000)	1.5e+04
Other transportation equipment	−3.155	7.083	1694	13.232 (0.000)	264.558
Electronics and electrical appliances	−0.029**	0.011	1812	10.912 (0.001)	613.420
Other manufacturing	−0.168	1.184	1812	17.254 (0.000)	1755.553
Weighted across all industries	−2.101	1.375	1838	5.409 (0.020)	9.727

Note: Since 'electronics, electrical appliances, and instrumentation manufacturing' accounts for a large share of China's exports, even a small estimated coefficient implies substantial fluctuations in export volume. The values in parentheses beside the KP rk LM statistics are the corresponding *p*-values, and the critical values for the Stock-Yogo weak instrument test at the 10% level corresponding to the KP rk WF statistic are all '16.380'. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

TABLE 14 | Expanding new export markets.

	(1)	(2)	(3)	(4)
Variables	Exports to competitive countries	Exports to competitive countries	Exports to other countries worldwide	Exports to other countries worldwide
<i>ETFRM</i>	−4.254*** (1.494)	−3.959*** (1.413)	−4.120*** (1.503)	−4.095*** (1.440)
<i>ETDR</i>	1.714*** (0.472)	1.730*** (0.483)	1.412*** (0.423)	1.504*** (0.438)
<i>ETFRC</i>	−5.080** (2.508)	−4.614 (2.808)	−3.883 (2.375)	−3.510 (2.921)
City controls	Yes	Yes	Yes	Yes
Main effects	Yes	Yes	Yes	Yes
<i>KP rk LM</i>	17.584 (0.000)	17.435 (0.000)	17.584 (0.000)	17.435 (0.000)
<i>KP rk WF</i>	64.761 (16.380)	79.403 (16.380)	64.761 (16.380)	79.403 (16.380)
City FE	Yes	No	Yes	No
City × year FE	No	Yes	No	Yes
Industry × year FE	Yes	Yes	Yes	Yes
Observations number	19,842	21,616	19,842	21,616
<i>R</i> ²	0.020	0.010	0.022	0.009

Note: *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

The underlying reason for this phenomenon may lie in the production characteristics of different industries and the varying levels of robot adoption. In labour-intensive industries such as

apparel and textiles, although robots have a substitution effect, developed countries have relatively low robot adoption in these industries, leading to a weaker substitution effect.

In industries such as chemicals and chemical engineering, metal machinery, and automobile manufacturing, while developed countries widely adopt robots, these industries inherently employ fewer workers, making it difficult for the robot substitution effect to manifest. In contrast, in the electronics and electrical equipment industry, developed countries have invested heavily in robotics, while China primarily occupies downstream positions in the GVC, such as final product assembly, which is highly routinised and labour-intensive. As a result, the robot substitution effect is more pronounced. Based on this, the industry-specific regression results show that only China's electronics and electrical equipment industry experiences a significant negative impact on exports. Although the regression coefficients for apparel and textiles, chemicals and chemical engineering, and automobile manufacturing are also negative, the effects are not statistically significant.

5.3.2 | Expanding New Export Markets

When developed market-oriented countries adopt robots, China faces a 're-offshoring' shock, which reduces its trade volume with these countries. In this context, China actively seeks to develop new customers. This section examines whether the use of robots in developed countries impacts China's exports to other countries. Using customs data, we calculate China's 'city-industry' level exports to competitive countries and other countries from 2010 to 2016 and use this as the dependent variable. The core endogenous explanatory variable is the penetration rate of robot usage in developed countries, and corresponding instrumental variables are introduced for two-stage least squares estimation. The estimation controls for main effects, city time-varying variables, and fixed effects for city, industry, and year. For robustness checks, we replace the fixed effects with 'city-year' interaction fixed effects and 'industry-year' interaction fixed effects to strictly control for interference factors.

The estimation results are presented in Table 14. It can be observed that with the increase in robot usage penetration in developed countries, the export value from China to competitive countries and other countries around the world has declined. This finding indicates that the use of robots in developed countries also hinders China's efforts to expand into new export markets. One possible reason is that as Chinese labour costs rise, developed countries may use robots or outsource to other countries with lower labour costs, thereby creating new competitors for Chinese exports in the international market, which impedes the development of new export markets.

6 | Conclusion

Industrial robots are widely adopted globally, substituting labour with capital, which adjusts the relative costs of production between countries and shifts the direction of international trade. This article combines a task model with a multi-sector Ricardian model and constructs an index to measure the impact of robot usage in developed market-oriented countries on China. Both theoretical and empirical analyses are conducted to

investigate whether developed market-oriented countries, represented by Europe and the United States, exert a shock on China's exports following their adoption of industrial robots.

The motivation for developed market-oriented countries to adopt industrial robots stems from the growing maturity of automation technology globally and the gradual disappearance of cheap labour. Within the global division of labour network, the impact of rising production costs in China will spill over through cross-country and cross-industry input-output trade. For developed countries that utilise robots to automate traditional industries, the relative production costs between countries are reversed. After accounting for the cost changes in various countries, this study concludes that the substitution of Chinese labour by robots in developed countries will reduce China's exports. Additionally, when similar emerging countries adopt robots, the impact on China's exports is smaller, as industrial complementarity outweighs competition between countries. These findings indicate that while China's industrial structure is continuously upgrading, it has not yet reached the level of developed countries and remains susceptible to shocks from them.

This article also identifies industry heterogeneity in the impact of foreign robot usage. On one hand, industries with a higher proportion of simple tasks have more tasks that can be replaced by robots, thus experiencing a stronger shock. On the other hand, industries that require more intermediate inputs from other sectors for production, as well as those with greater trade facilitation with foreign countries, are more severely impacted. Finally, the study also finds that China's concurrent increase in robot usage can alleviate labour cost pressures, thereby partially offsetting the negative impact of robot adoption in developed countries.

However, this article has some limitations that require further research in the future. In terms of the model, due to the involvement of intermediate goods production cost changes at the 'country-industry' level within the network effects of the GVC, the impact on trade is extremely complex. Consequently, this article had to make some realistic assumptions to complete the analysis. Additionally, the model does not capture the productivity-enhancing effects of industrial robots or consider the impact of scale expansion on 're-offshoring'. In terms of empirical analysis, the lack of more detailed regional and industry-level foreign economic data in China means that there is still room for improvement in measuring the domestic penetration rate of robot usage in developed countries.

Funding

This work was supported by General Program of the National Natural Science Foundation of China, 72273068; Key Program of the National Natural Science Foundation of China, 72433002; Liaoning Provincial Department of Education University Basic Scientific Research Projects Publicity, Zong20250169 and Major Programme of the National Social Science Fund of China, 22&ZD097.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Endnotes

- ¹ In the traditional outsourcing model, developed countries and regions such as Europe, the United States, and Japan transferred low-tech, low-value-added downstream and midstream industries to China for production, while China exported finished goods to developed country markets. In this article, these countries are referred to as 'developed market-oriented countries', or simply 'developed countries'.
- ² The specific factor can be considered as an input of raw materials, which will be set to zero in the subsequent analysis.
- ³ This implies that the specific factor required to complete tasks is instead represented by intermediate input in production.
- ⁴ The 2010 China Industrial Economic Statistical Yearbook lacks national employment data for 10 industries, including leather, fur, and feather products and footwear, wood processing and products made of bamboo, rattan, palm, and straw, furniture manufacturing, printing and recording media, among others. Additionally, the transportation equipment manufacturing sector is not classified. For the missing industries, this paper estimates the total employment in 2010 by using the reported employment growth rates for each industry from the 2013 to 2016 China Industrial Economic Statistical Yearbooks. For the transportation equipment manufacturing sector, the study separates the two sub-sectors based on proportions calculated from the 2010 Population Census.
- ⁵ In 2010, there were 9,744,454 export transactions. The total trade value was aggregated by developed countries and other emerging countries, and the corresponding ratio f was calculated.
- ⁶ The Chinese National Population Census surveys residents' current place of residence and employment industry information. This paper uses a 2010 Population Census sample dataset, which contains 4,400,367 observations and is nationally representative. Using this data, the ratio of employment in industry i within city c to the total employment in city c is calculated, providing a more accurate estimate. Additionally, in 2010, the 2002 version of the GB classification was still in use, categorising the automotive manufacturing industry and other transportation equipment manufacturing, including rail, shipbuilding, and aerospace, under the broader category of transportation equipment manufacturing. In this paper, category 372 within the transportation equipment manufacturing subsector is designated as the automotive manufacturing industry, while the remaining categories 371–379 are classified as other transportation equipment manufacturing.
- ⁷ The customs data consists of all export transaction records documented by Chinese customs throughout the year, providing information such as HS 8-digit codes, the location of enterprises, destination countries, and export values. In this paper, the HS codes are matched with IFR industries to determine the industry for each export transaction. The enterprise locations are matched with the city administrative codes released by the Ministry of Civil Affairs in 2019 to identify the city of each export. Additionally, the destination country can be determined through the country code, allowing for the aggregation of export data at the city-industry level. For exports from different years, the data are deflated using the export price deflator.
- ⁸ The minimum wage standard at the district and county level is sourced from CNRDS, and this paper selects the highest minimum wage level within a city as the city's minimum wage level.
- ⁹ Faber (2020) used city-level data, which made it challenging to control for time-varying disturbances during the study. Therefore, he employed a method of mean testing for observable variables. This paper constructs two-dimensional city-industry data, allowing for

the control of unobservable factors from multiple dimensions. This approach maximises the independence of the components of the instrumental variables from the random error term, ensuring the accuracy of the estimation results.

- ¹⁰ Due to the absence of Swiss data in the WIOD, only 13 market-oriented countries are included here.
- ¹¹ This relates to the issue of industry positioning. If a developed market-oriented country n locates core processes, such as R&D, for industry j domestically while assigning simple processing and assembly stages to developing country i , then industry j in country n will have a higher position in the global value chain, while the position of country i will be relatively lower.
- ¹² The 2015 and 2016 samples in the dataset correspond to the 2014 WIOD data.
- ¹³ <https://unstats.un.org/unsd/trade/classifications/correspondence-tables.asp>.

References

- Acemoglu, D., and P. Restrepo. 2018. "The Race Between Man and Machine: Implications of Technology for Growth, Factor Shares, and Employment." *American Economic Review* 108, no. 6: 1488–1542.
- Acemoglu, D., and P. Restrepo. 2020. "Robots and Jobs: Evidence From US Labor Markets." *Journal of Political Economy* 128, no. 6: 2188–2244.
- Acemoglu, D., and F. Zilibotti. 2001. "Productivity Difference." *Quarterly Journal of Economics* 116: 563–606.
- Artuc, E., P. Bastos, and B. Rijkers. 2023. "Robots, Tasks, and Trade." *Journal of International Economics* 145: 103828.
- Caliendo, L., and F. Parro. 2015. "Estimates of the Trade and Welfare Effects of NAFTA." *Review of Economics Studies* 82: 1–44.
- Eaton, J., and S. S. Kortum. 2002. "Technology, Geography, and Trade." *Econometrica* 70: 1741–1779.
- Faber, M. 2020. "Robots and Reshoring: Evidence From Mexican Labor Markets." *Journal of International Economics* 127: 1–34.
- Frey, C. B., and M. A. Osborne. 2017. "The Future of Employment: How Susceptible Are Jobs to Computerization?" *Technological Forecasting and Social Change* 114: 254–280.
- Gihleb, R., O. Giuntella, L. Stella, and T. Wang. 2022. "Industrial Robots, Workers' Safety, and Health." NBER Working Paper.
- Goldsmith-Pinkham, P., I. Sorkin, and H. Swift. 2020. "Bartik Instruments: What, When, Why, and How." *American Economic Review* 110, no. 8: 2586–2624.
- Graetz, G., and G. Michaels. 2018. "Robots at Work." *Review of Economics and Statistics* 100, no. 5: 753–768.
- Grossman, M. G., and E. Rossi-Hansberg. 2008. "Trading Tasks: A Simple Theory of Offshoring." *American Economic Review* 98, no. 5: 1978–1997.
- Humlum, A. 2019. "Robot Adoption and Labor Market Dynamics." Working Paper.
- IFR. 2019. *World Robotics 2019- Industrial Robots*. IFR.
- Koopman, R., W. Powers, Z. Wang, and S. J. Wei. 2010. "Give Credit Where Credit Is due: Tracing Value Added in Global Production Chains." NBER Working Paper.
- Krenz, A., K. Prettnner, and H. Strulik. 2021. "Robots, Reshoring, and the Lot of Low-Skilled Workers." *European Economic Review* 136: 103744.
- Ma, H., L. Li, B. Sheng, and G. Xu. 2024. "Impact of Global Production Automation on China's Trade and Welfare." *Journal of World Economy* 47, no. 9: 3–32.

Ofori, I. K., and S. A. Asongu. 2024. "Repackaging FDI for Inclusive Growth: Nullifying Effects and Policy-Relevant Thresholds of Governance." *Transnational Corporations Review* 16, no. 2: 200056.

Oyewo, B., A. Obanor, and C. Iwuanyanwu. 2023. "Determinants of the Adoption of Big Data Analytics in Business Consulting Service: A Survey of Multinational and Indigenous Consulting Firms." *Transnational Corporations Review* 15, no. 2: 1–20.

Sirkin, L. H., M. Zinser, and R. J. Rose. 2015. *The Robotics Revolution*. Boston Consulting Group.

Stemmler, H. 2019. "Does Automation Lead to De-Industrialization in Emerging Economics? Evidence From Brazil." CEGE Discussion Papers.